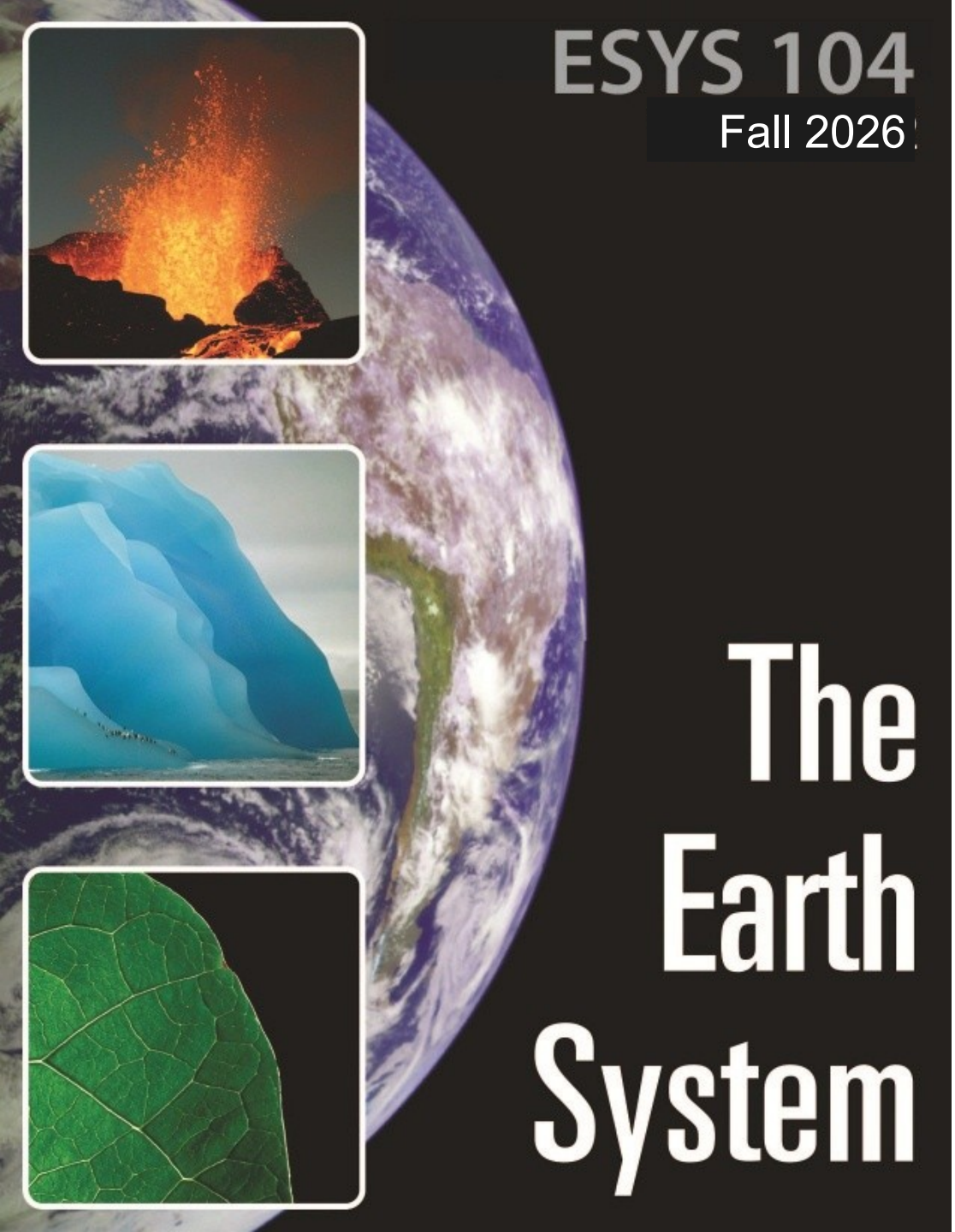


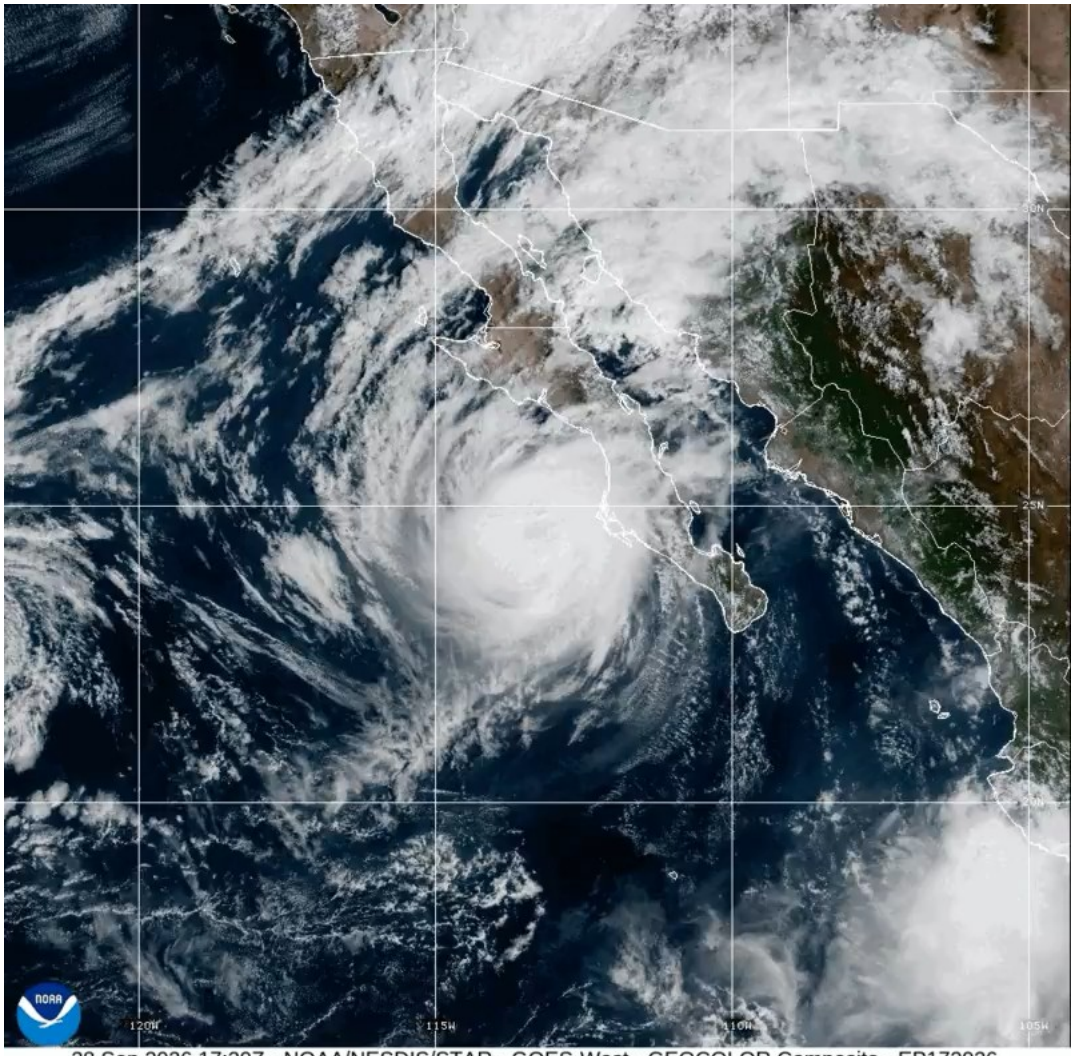
Intro to water cycle and soil fertility

Biomes and Ecosystems

Assignment #2 posted tonight

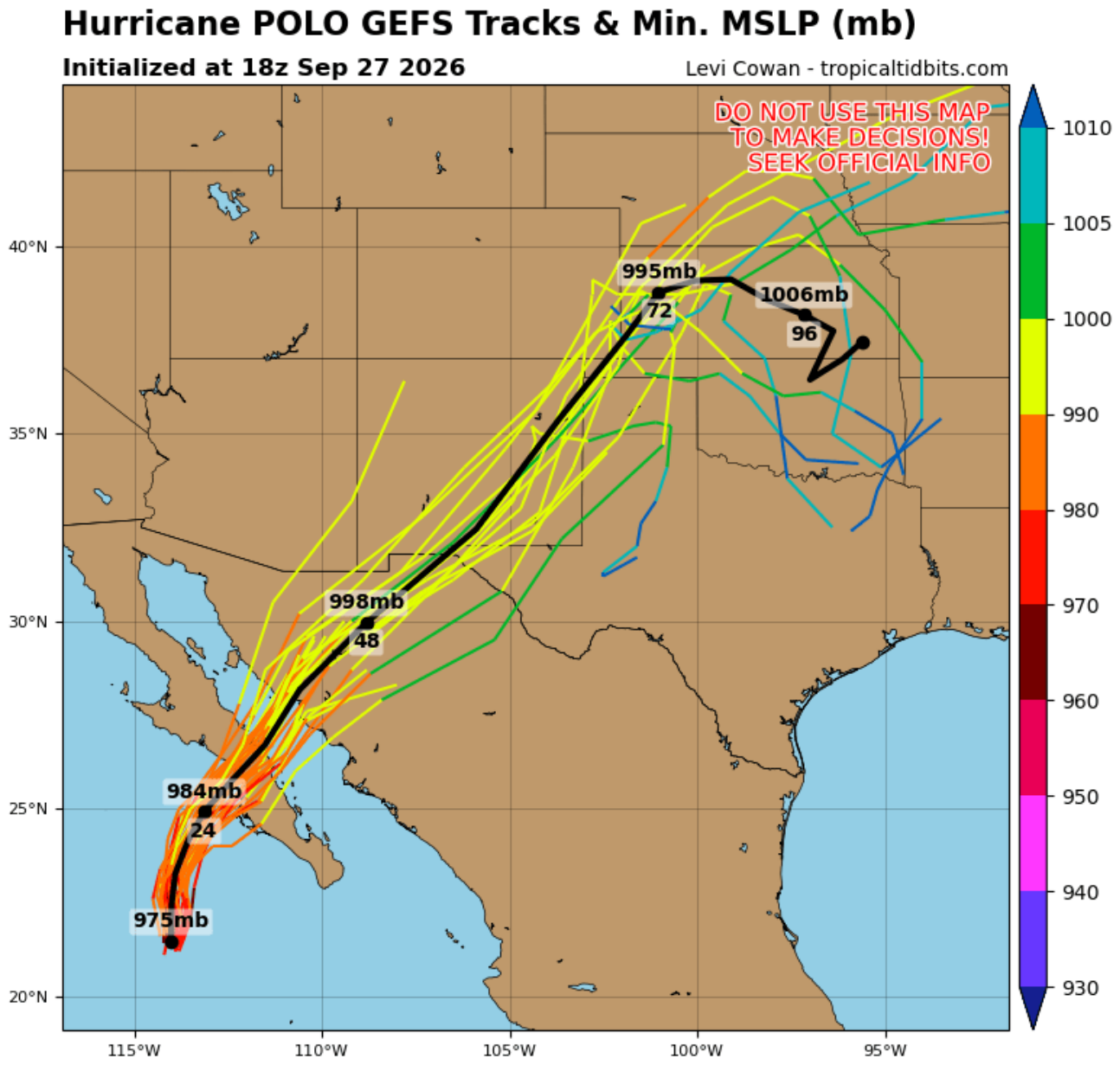
Quiz #2 today

Hurricane Polo

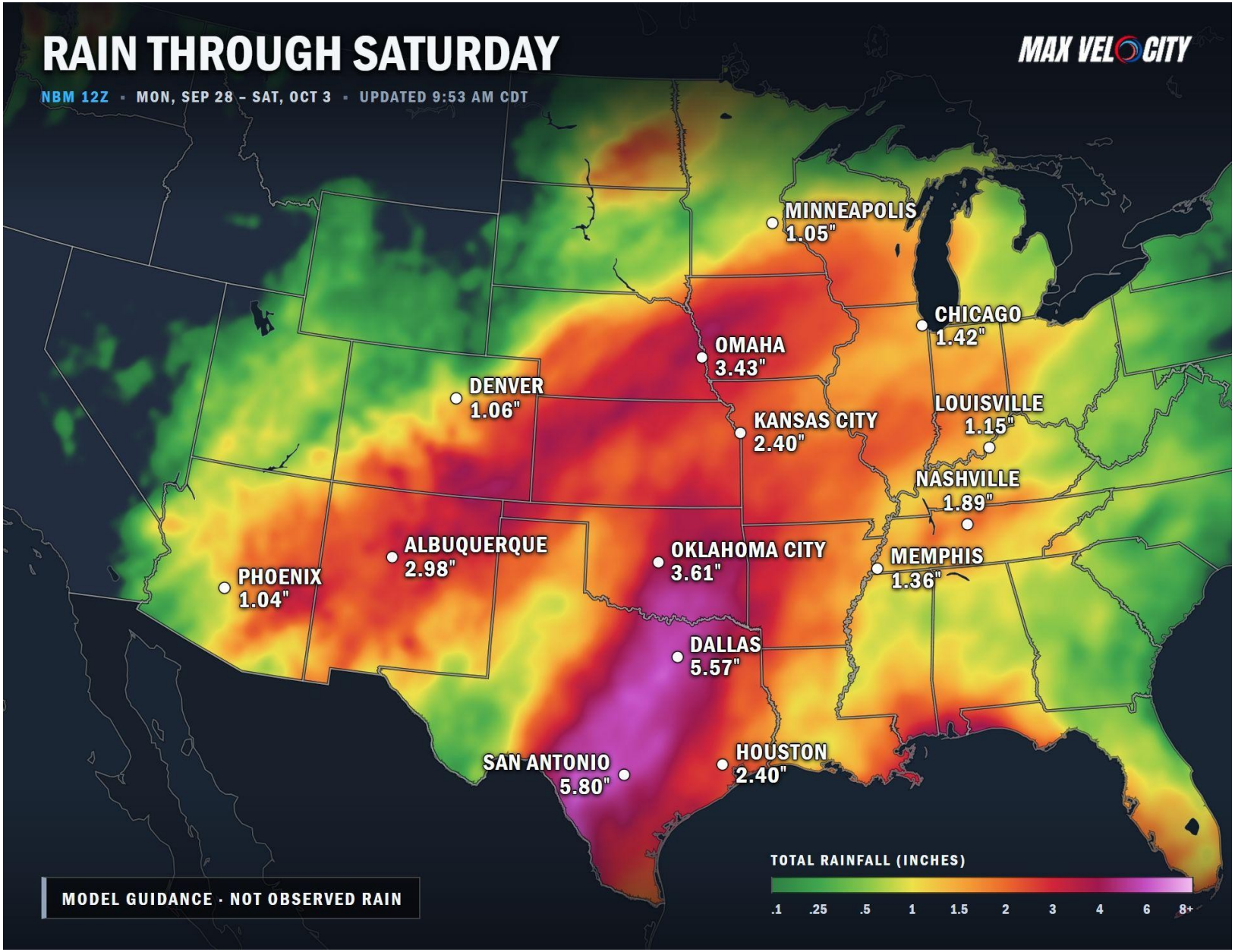


28 Sep 2026 17:20Z - NOAA/NESDIS/STAR - GOES-West - GEOCOLOR Composite - EP172026

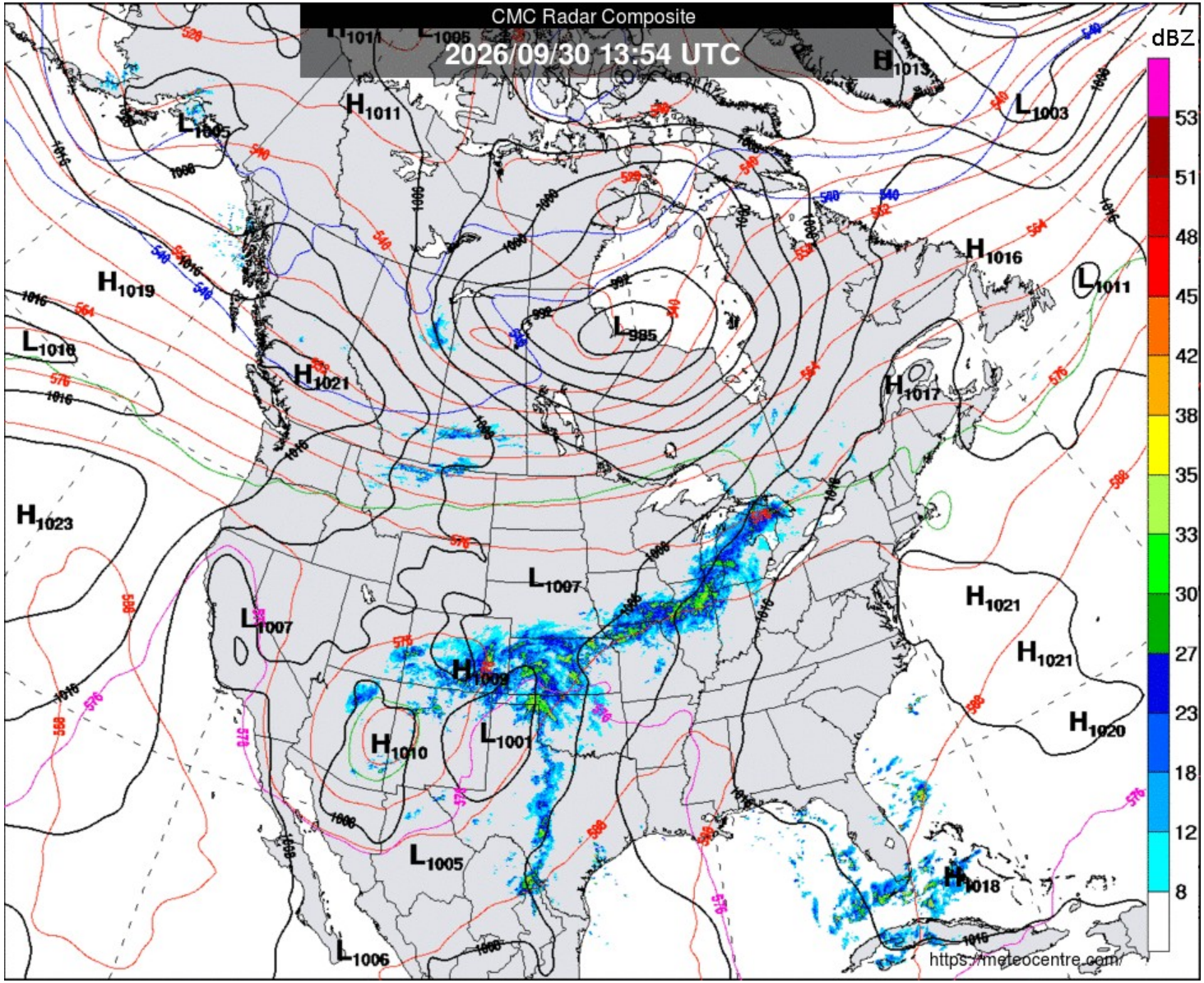
Hurricane Polo



Hurricane Polo



Hurricane Polo



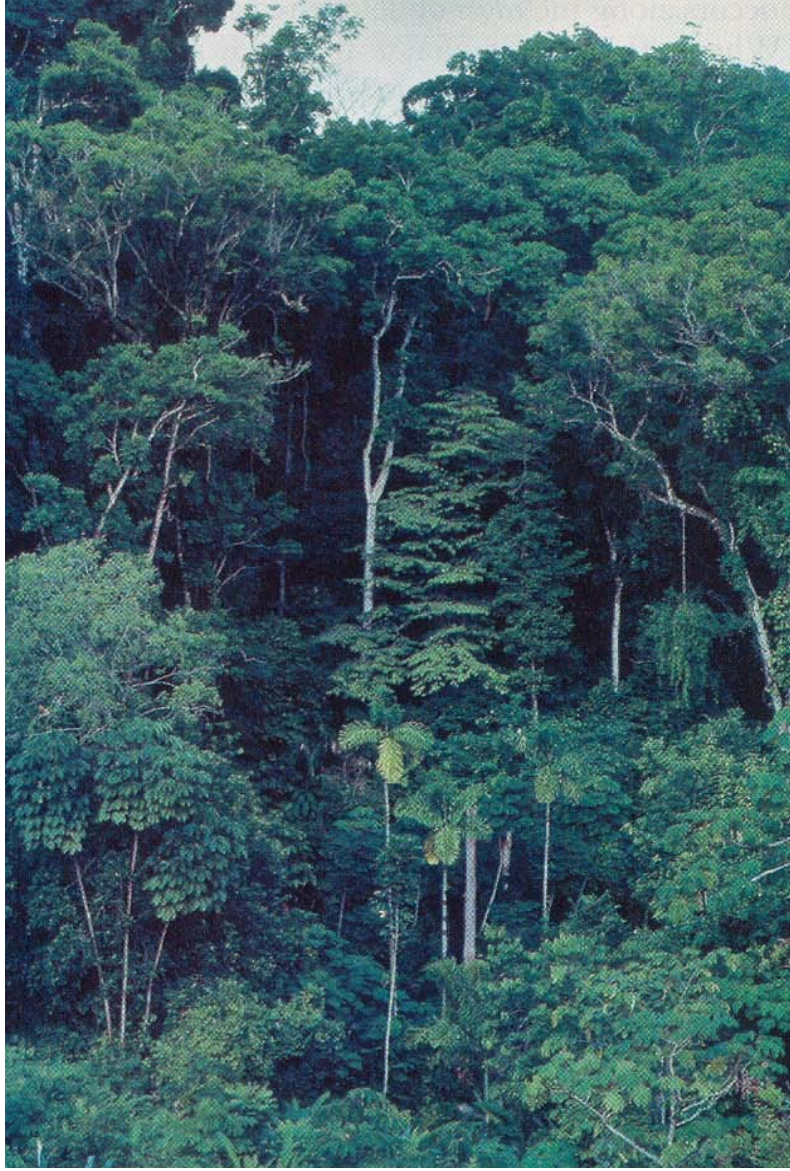
Today's Topics

The water balance equation
Groundwater

Biomes
    mean annual temperature and precipitation
Competition, Optimal Range, Plant Stress
Adaptations

Ecosystems
Food Web, Trophic Energy Pyramid
Ecosystem Services





Why this huge difference in natural vegetation cover (and ability to support humans and diverse types of animals) over various regions of Earth?



H_2O and soil moisture must be available for life



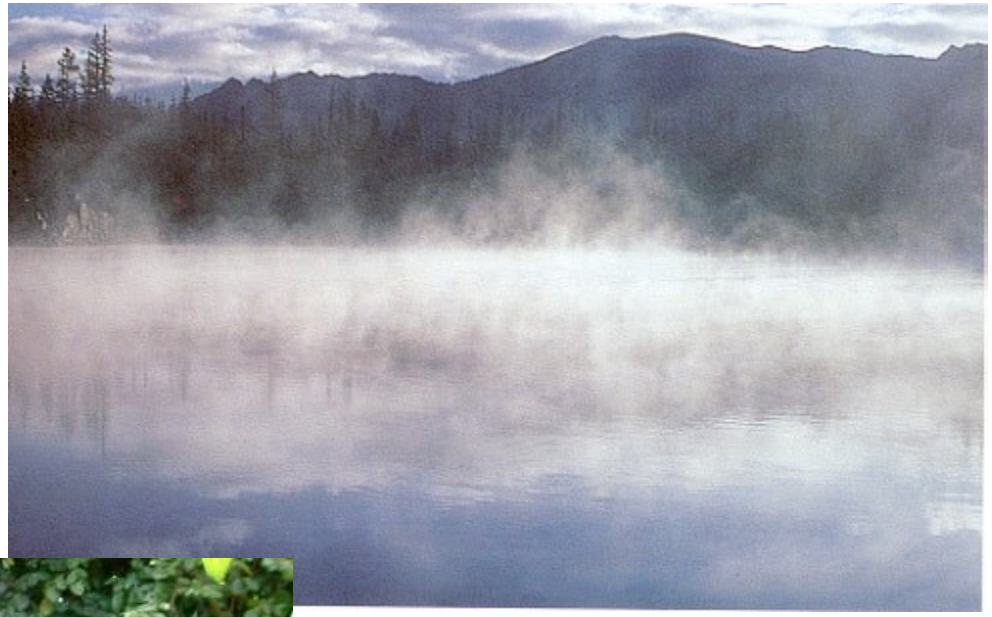


We will distinguish processes that determine how much water is available
in soils
in wells (groundwater)
in lakes and rivers

Intro to Hydrology and Soils
The Water Cycle, Soil Fertility

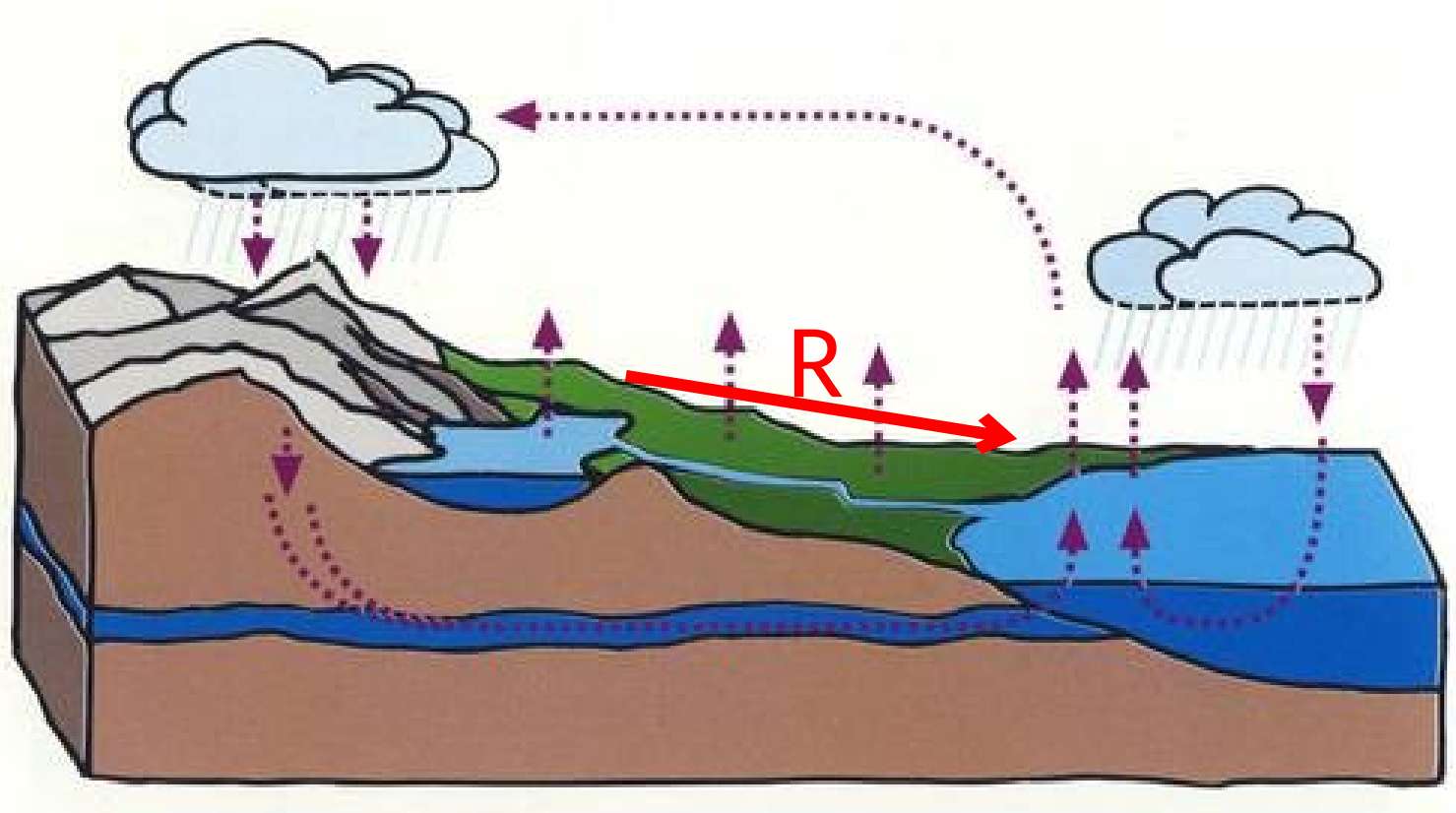
- The main components of the cycle
- The various water ‘stores’ on earth
- Soil water and infiltration, evapotranspiration
- The water budget equation for a ‘watershed’ runoff and groundwater
- Weathering of minerals and soil fertility

Evaporation: here, visible water “vapor” , rising above evaporating surface

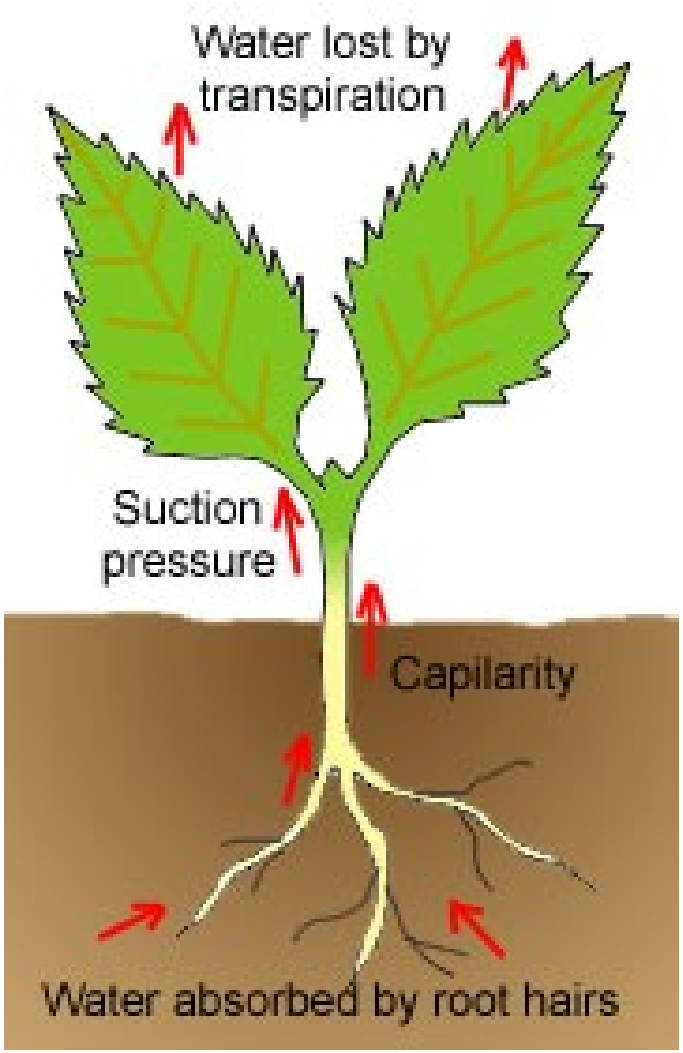




In the net, the water cycle transports water vapor from oceans and rains it onto lands, while Runoff R returns this excess water to oceans



Plant transpiration



Like evaporation from lakes and ponds , plant transpiration is also variable and controlled by climate (how warm and dry is the air, how strong is the sun , how strong is the wind near the ground, etc)

The two processes (pond evaporation and plant transpiration) are often combined as one vertical flux of H_2O vapor from land to air, called:

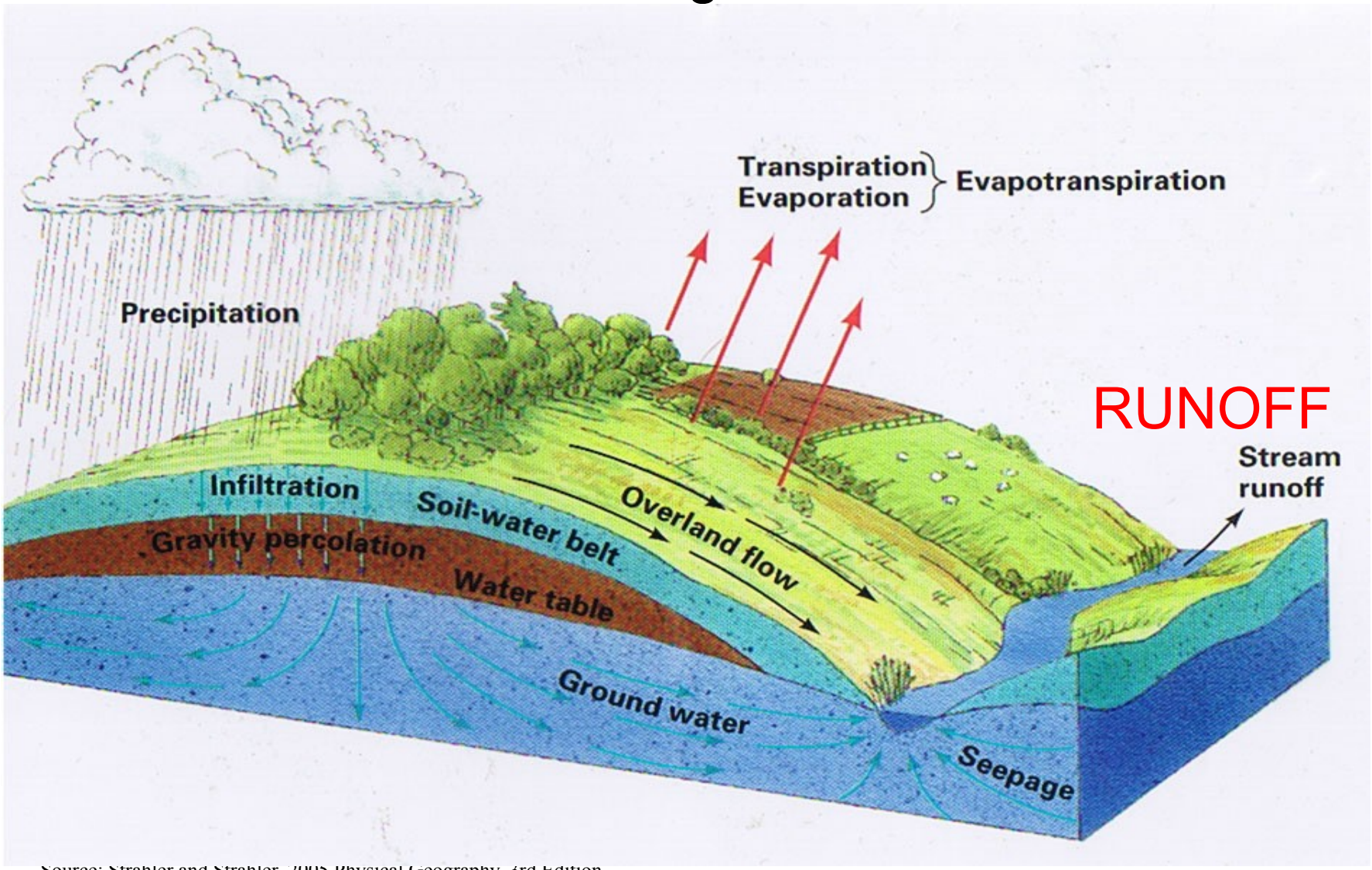
Evapo-transpiration (Et)

For plants to grow at their potential rate they need to transpire freely which requires that there always be enough water left in the soil. This "Potential Et" rate is also called "water demand"

Standard **evaporation pan:** used to measure variable "water demand", a function of the local "climate" (cloudiness, air T, air dryness, wind, hours of daylight)



The precipitation that has not evaporated back flows as river runoff via two routes: overland flow and groundwater flow



Source: Strahler and Strahler, 2005 Physical Geography, 3rd Edition

The water balance equation

Applied to an entire drainage basin over entire year:

$$P = E_t + R$$

(i.e. water inputs = outputs, averaged over 12 months)

(units of volume of water per year)

- P = annual precipitation.
- E_t = annual actual evapotranspiration.
- R = Runoff (flow of liquid river water from land via streams, available to use; includes groundwater).

Or: $R = P - E_t$

i.e. even if it rains, available water R is decreased if E_t is very high (in hot, dry climates)

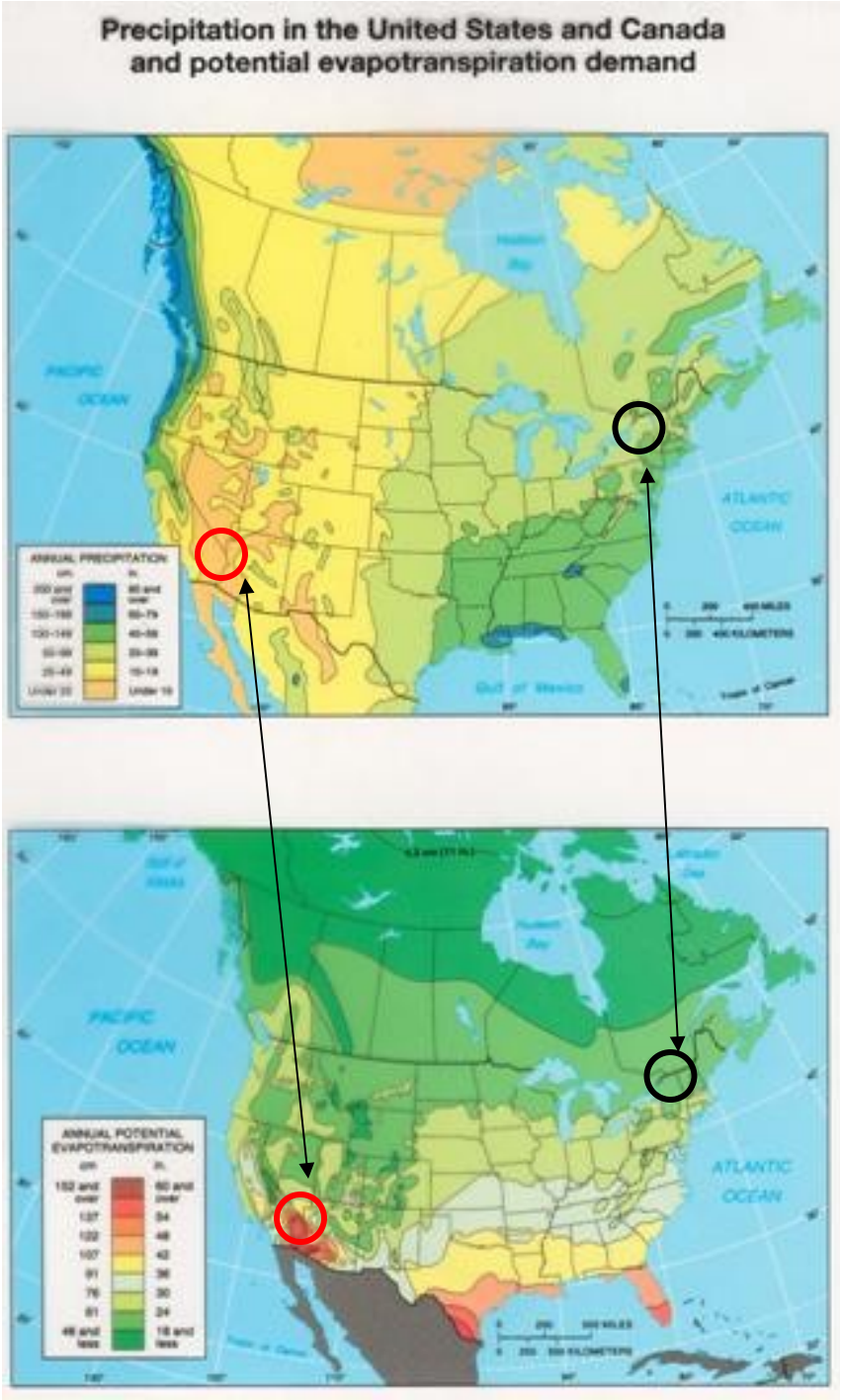
$$R = P - E_t$$

precipitation (top)

Maps contrast Montreal and ‘Death Valley’ (Southern California)

Which site has more runoff in rivers?

water demand: PE_t (bottom)



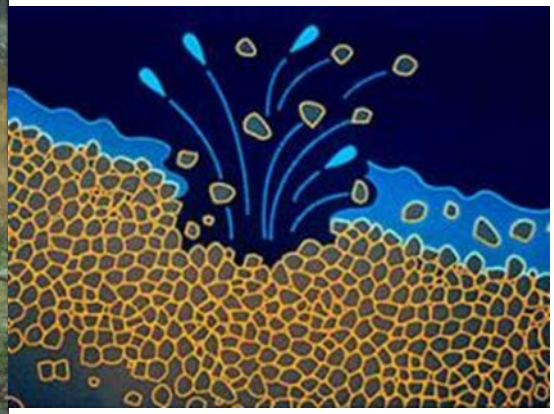
Christopherson, 2003, p 251-252.

Weir to measure runoff and stream chemistry



Rainsplash and overland flow on bare soil (e.g. on bare roads or farm fields after harvest) can be highly erosive (especially if there is a local slope)



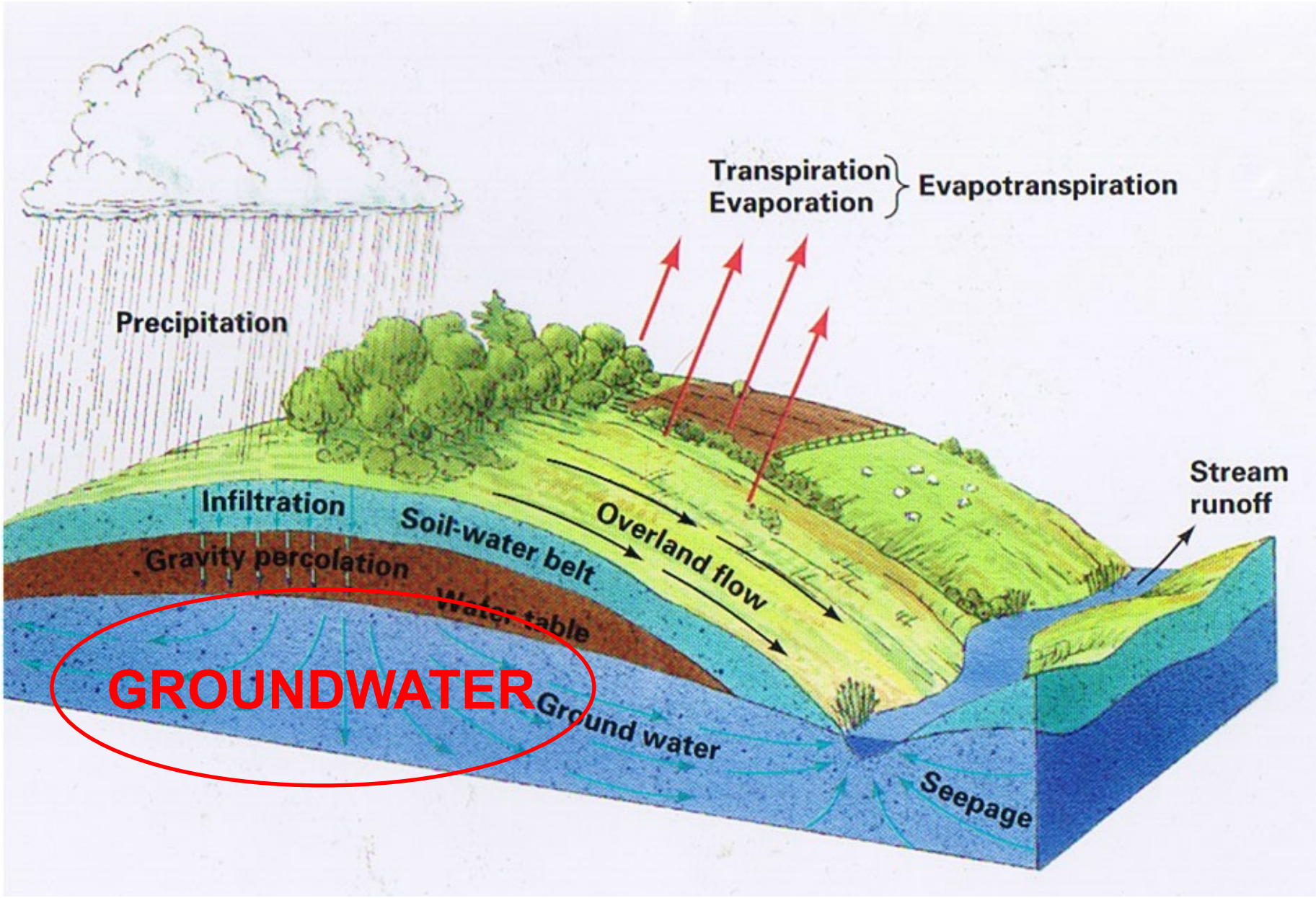


Strong showers over bare or cultivated lands can lead to erosive overland runoff and so to top-soil erosion (so excess fertilizer and pesticide runoff).



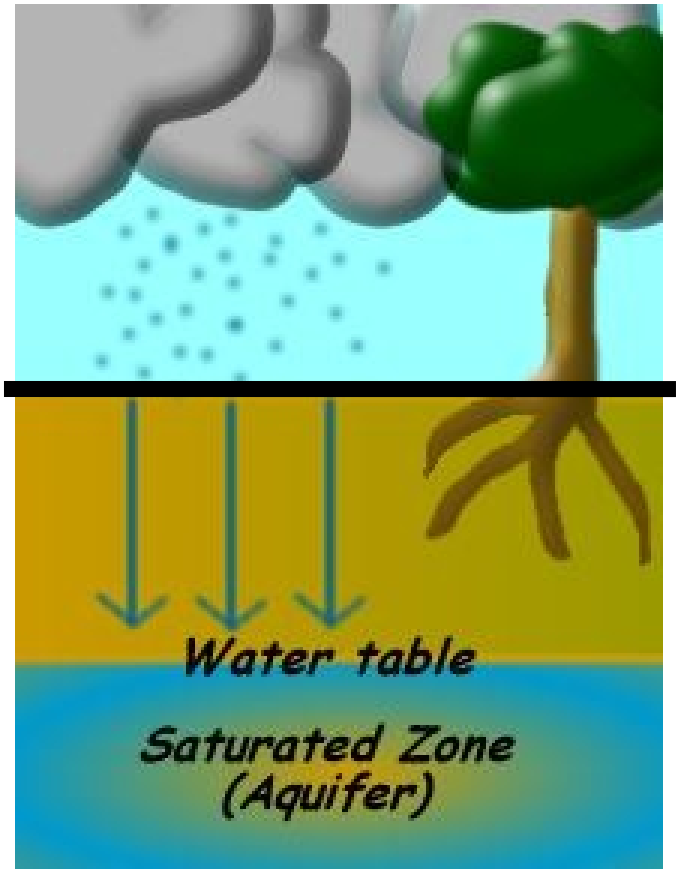


The hydrological cycle: the groundwater flow path



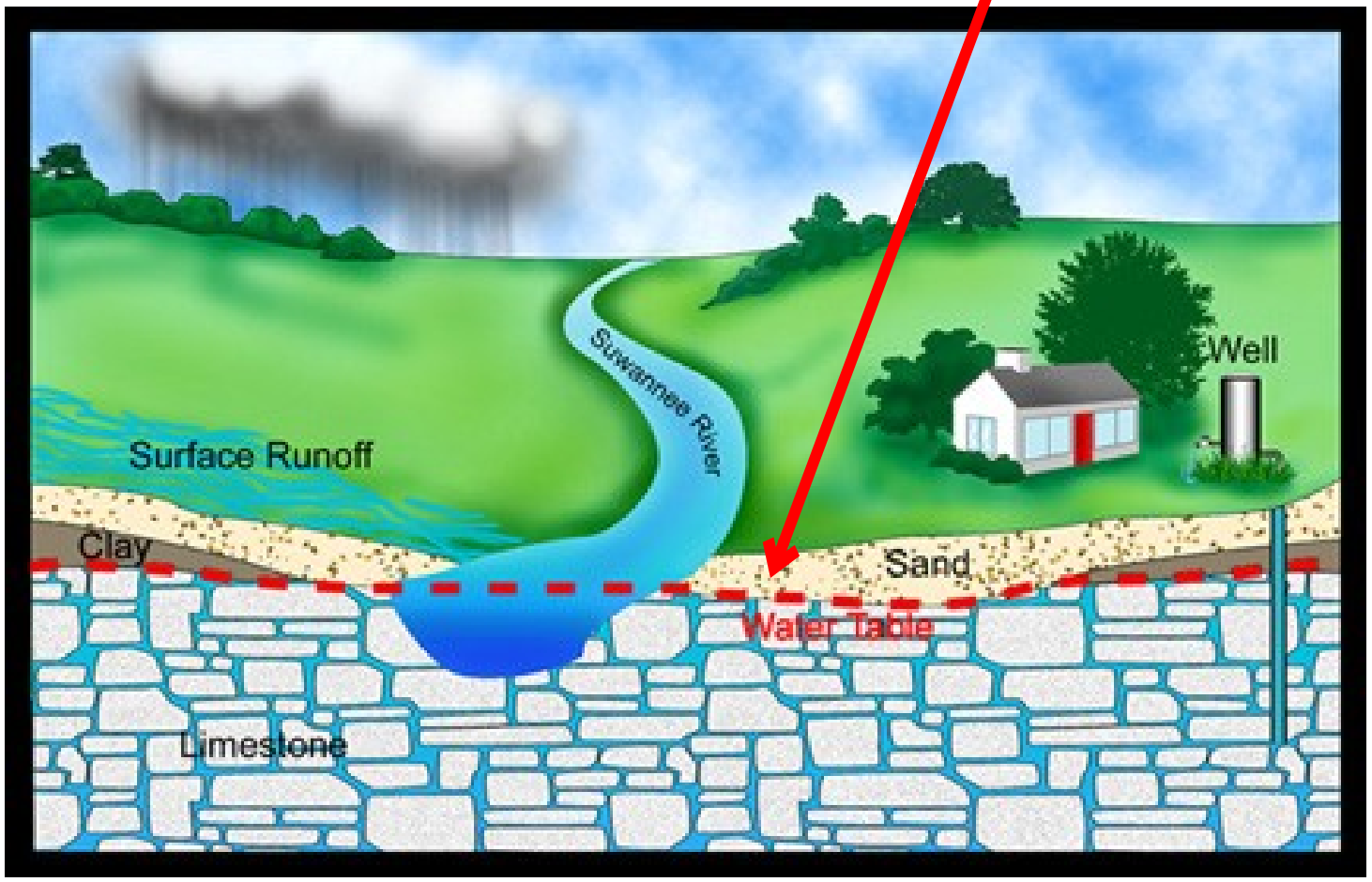
Source: Strahler and Strahler, 2005 Physical Geography, 3rd Edition

Recharge



Only the excess soil water, not transpired by plant roots, percolates down to the water table and adds to groundwater store in ‘aquifer’

The top of saturated zone is the “water table”



Groundwater flow velocities are variable but generally low

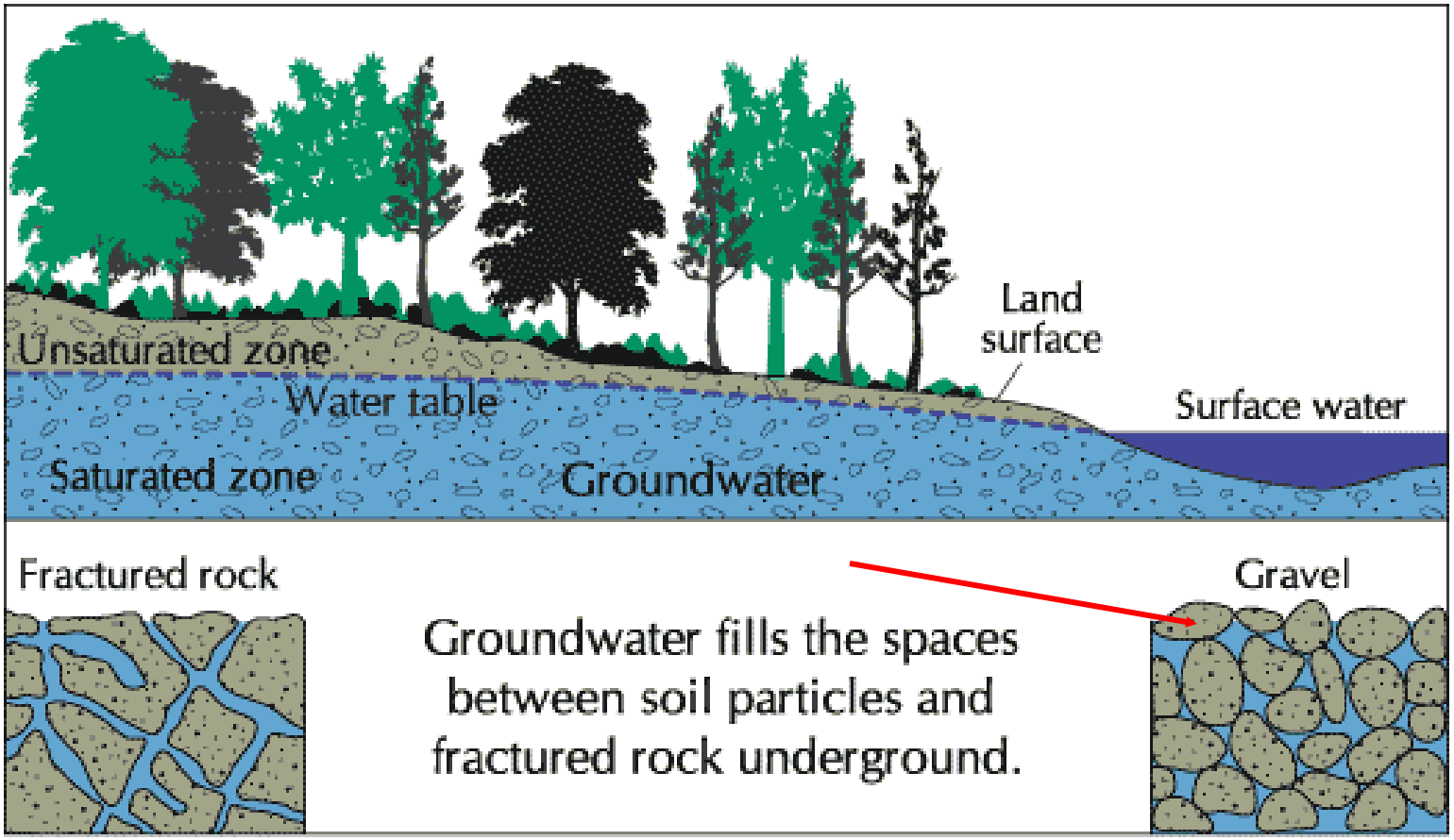
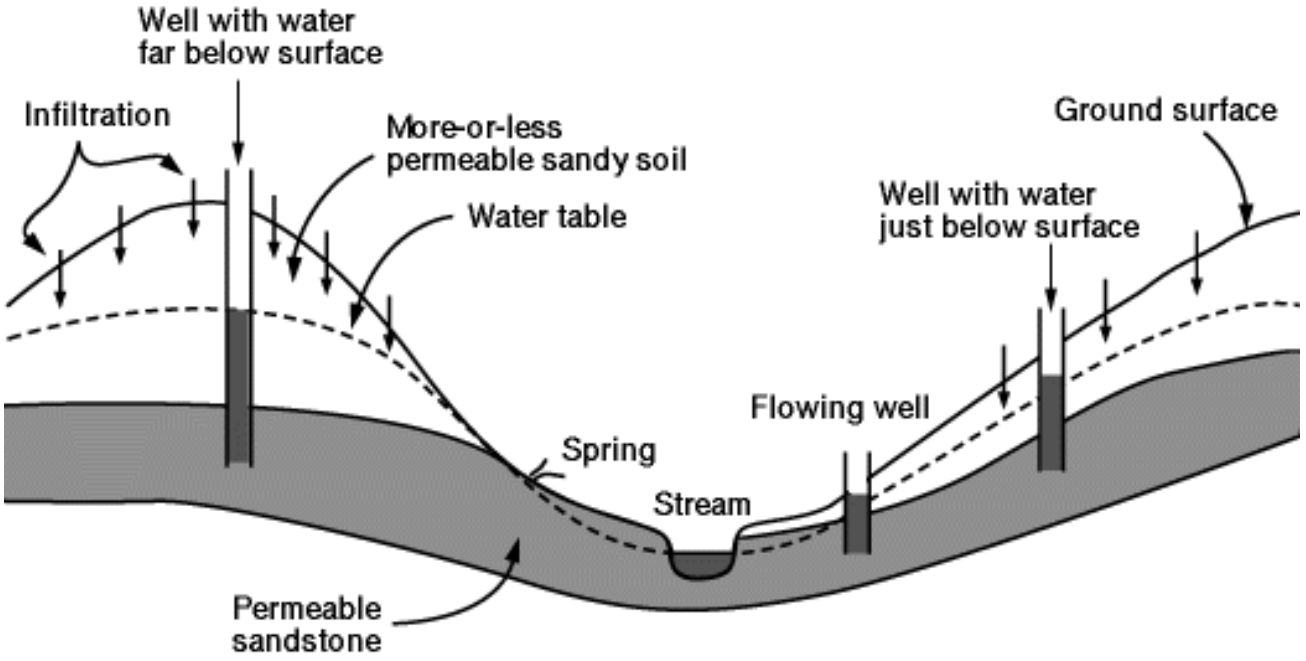


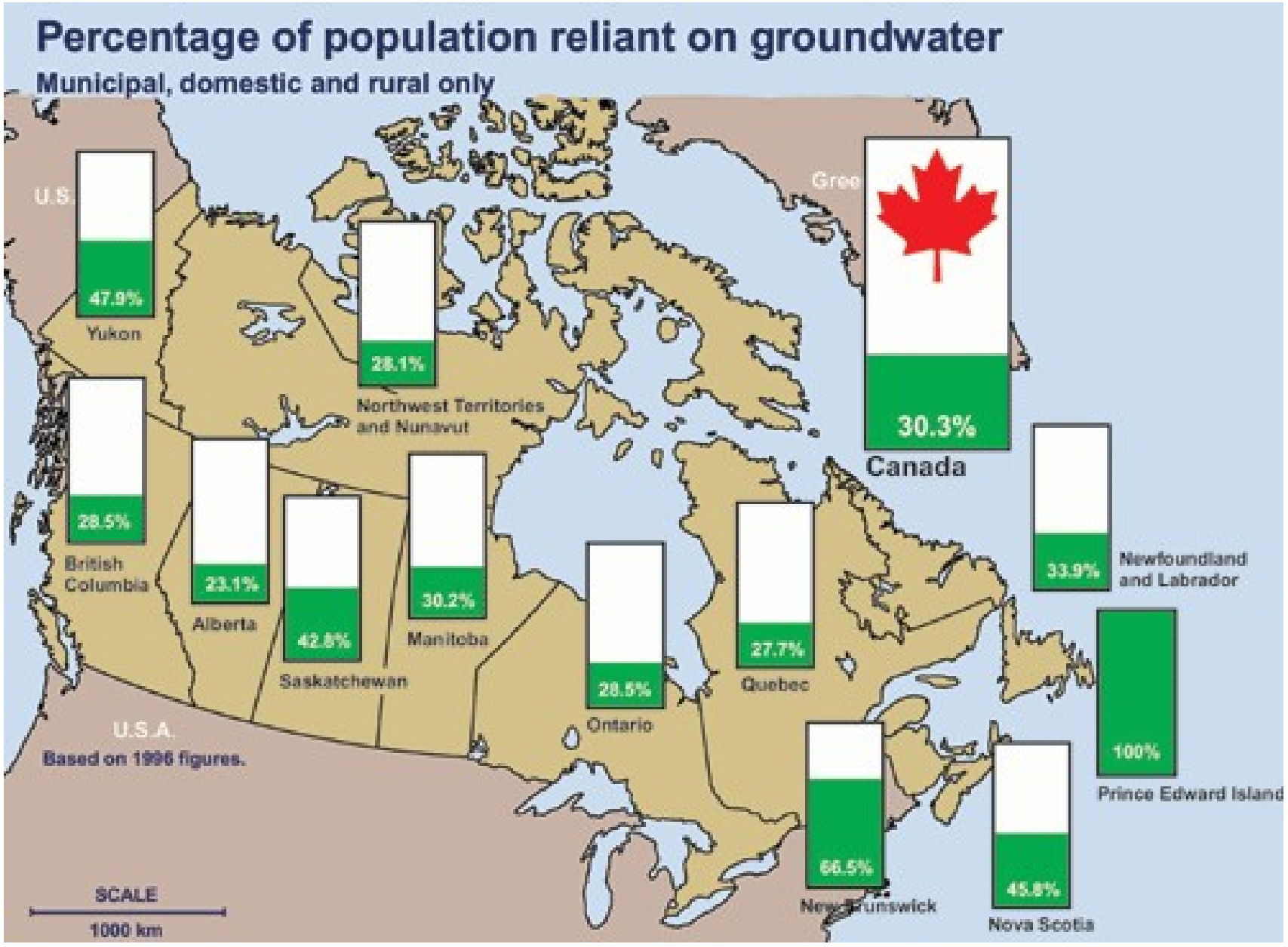
Image compliments of US Geological Survey, adapted by The Groundwater Foundation.



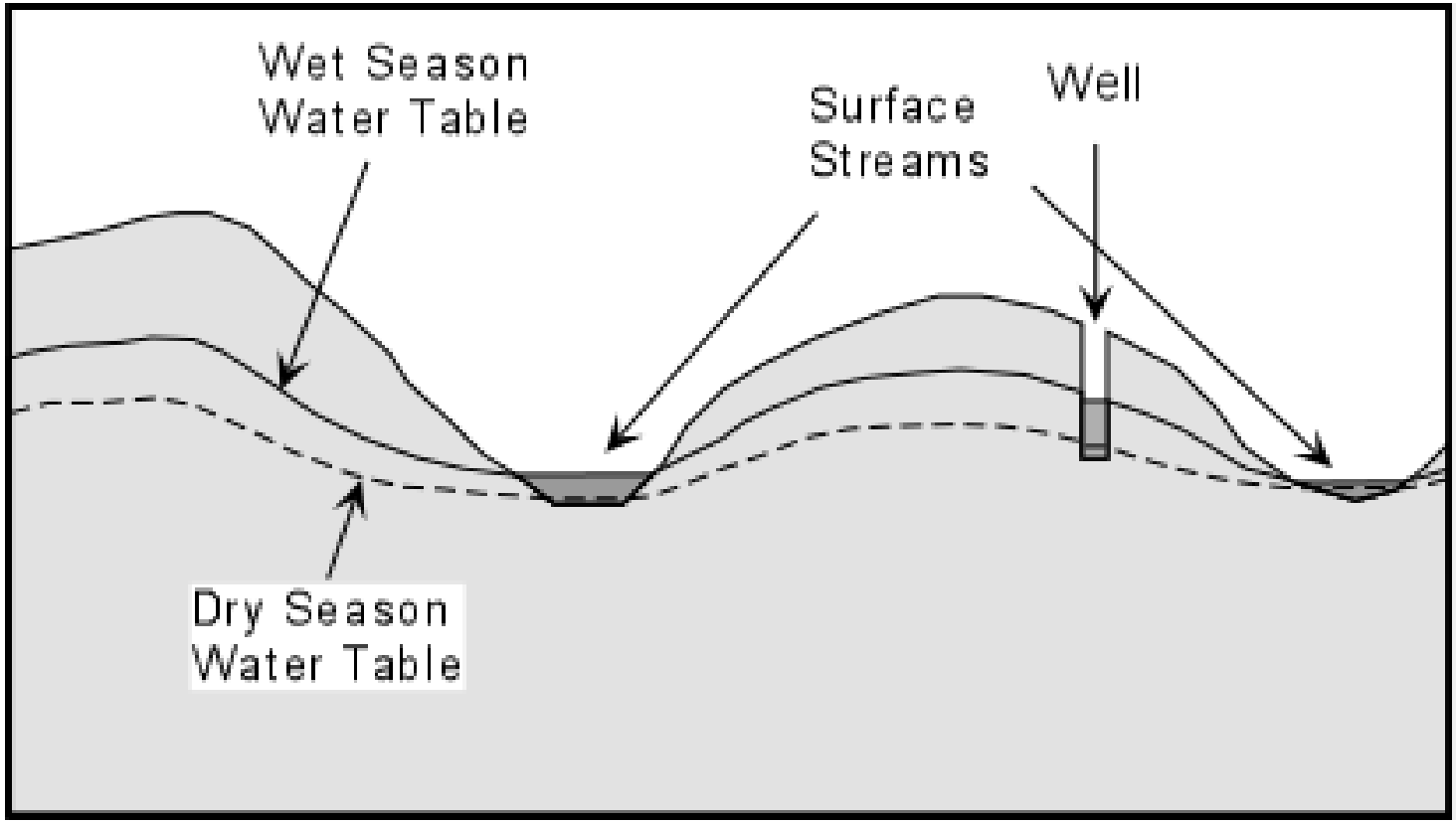
Water table goes up during wet seasons, and down during dry

In semi-arid climates (hot temperatures, low precipitation) where there are few streams and no lakes, wells provide the only abundant water during dry seasons on most farmsteads

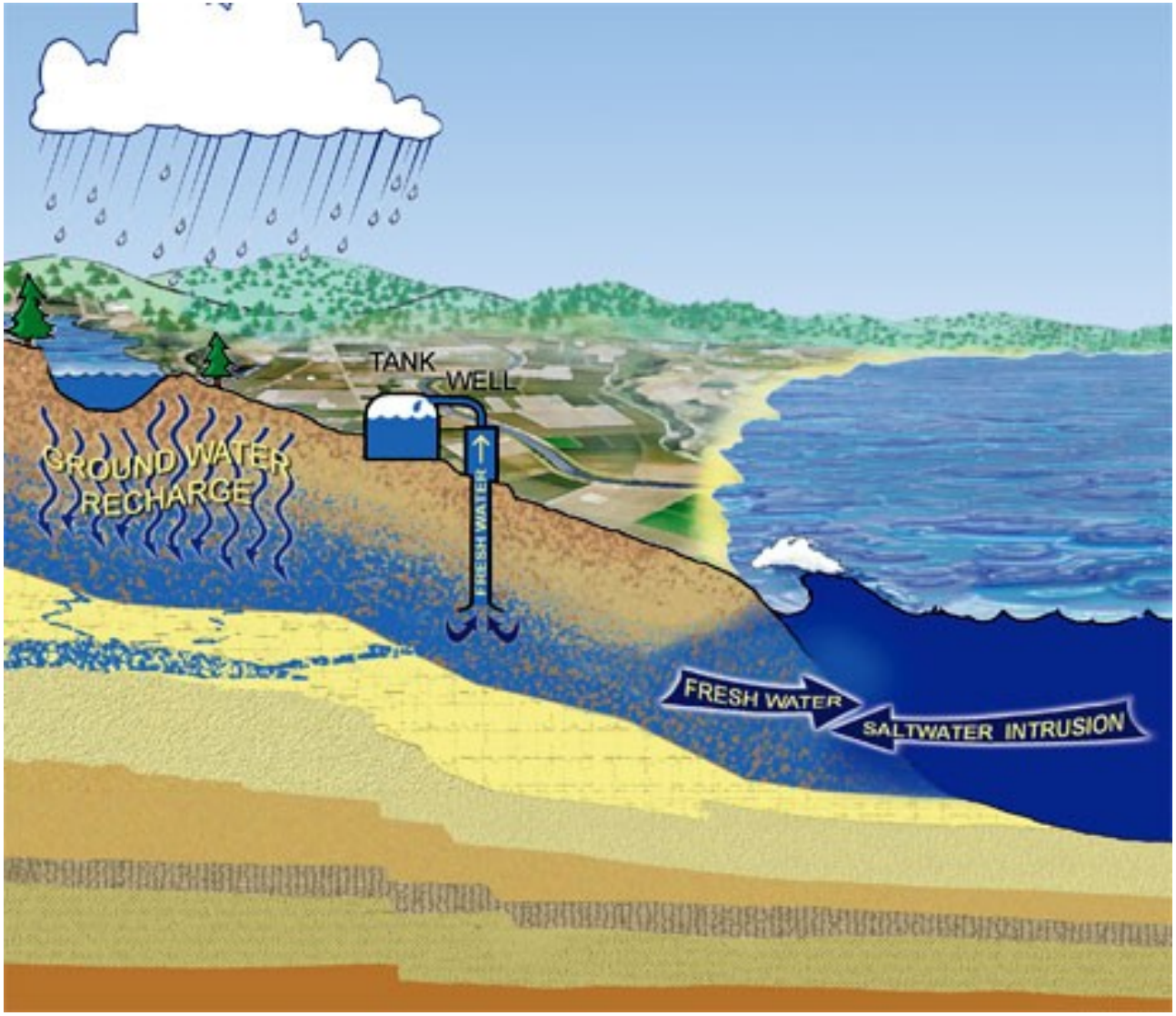



Percentage of population reliant on groundwater
Municipal, domestic and rural only
U.S.
47.9%
Yukon
28.1%
Northwest Territories and Nunavut
Gree
30.3%
Canada
28.5%
British Columbia
23.1%
Alberta
42.8%
Saskatchewan
30.2%
Manitoba
28.5%
Ontario
27.7%
Quebec
33.9%
Newfoundland and Labrador
100%
Prince Edward Island
66.5%
New Brunswick
45.8%
Nova Scotia
U.S.A.
Based on 1996 figures.
SCALE
1000 km

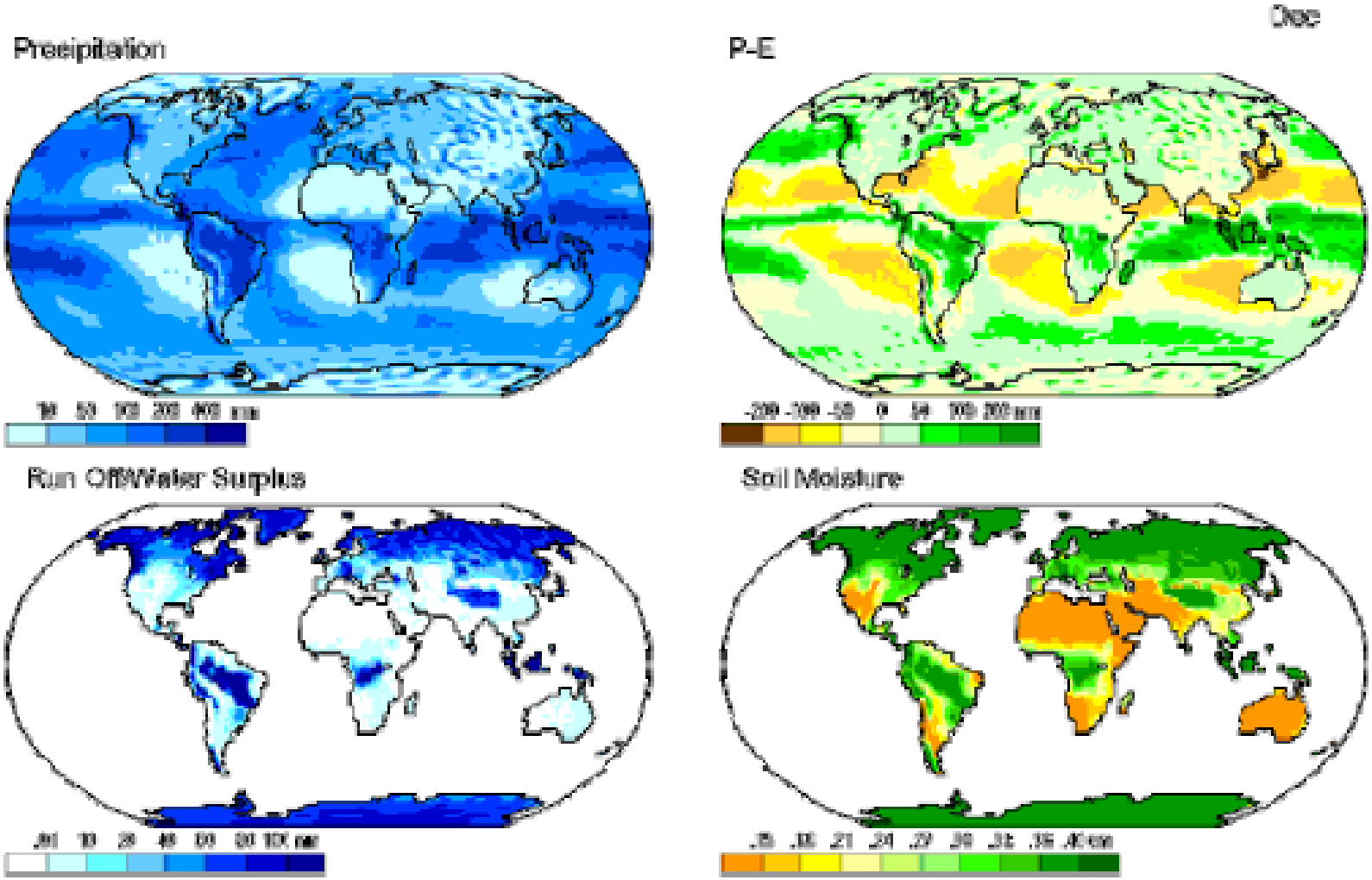
After long dry periods, water tables fall and groundwater outlow into streams can stop



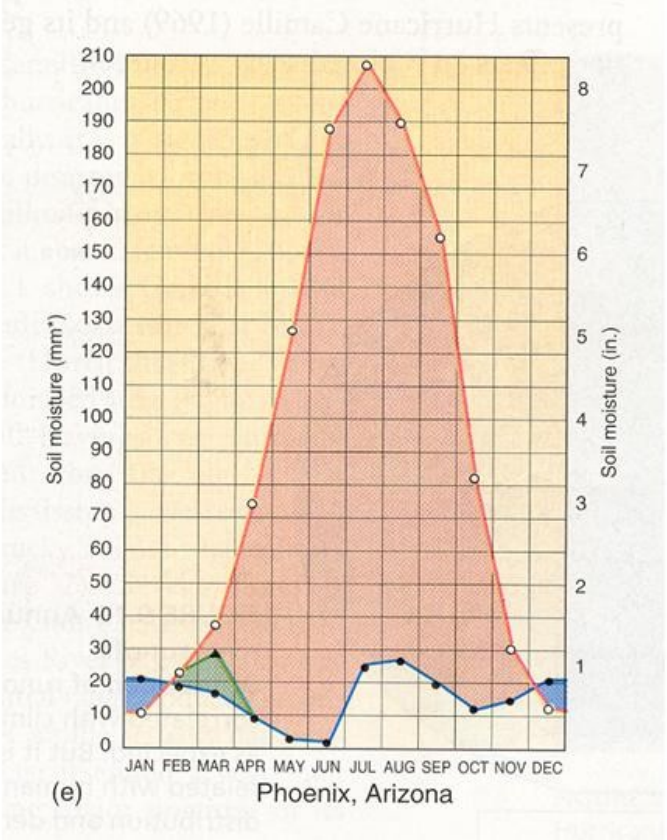
This can cause salt intrusion into fresh groundwater in some costal zones



Water Availability



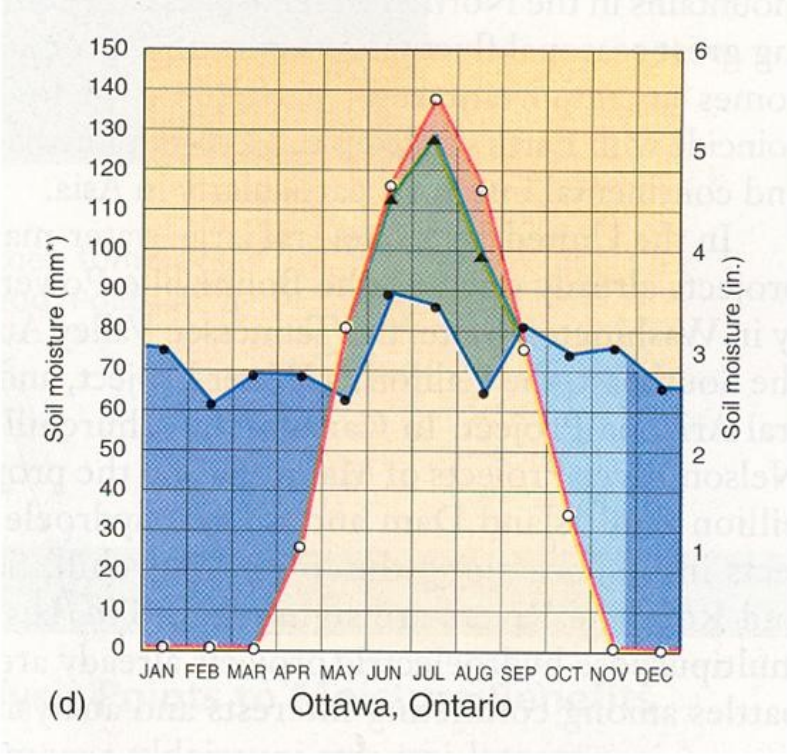
In many regions with soil moisture deficits (in yellow and brown, at bottom right) agriculture depends on irrigation



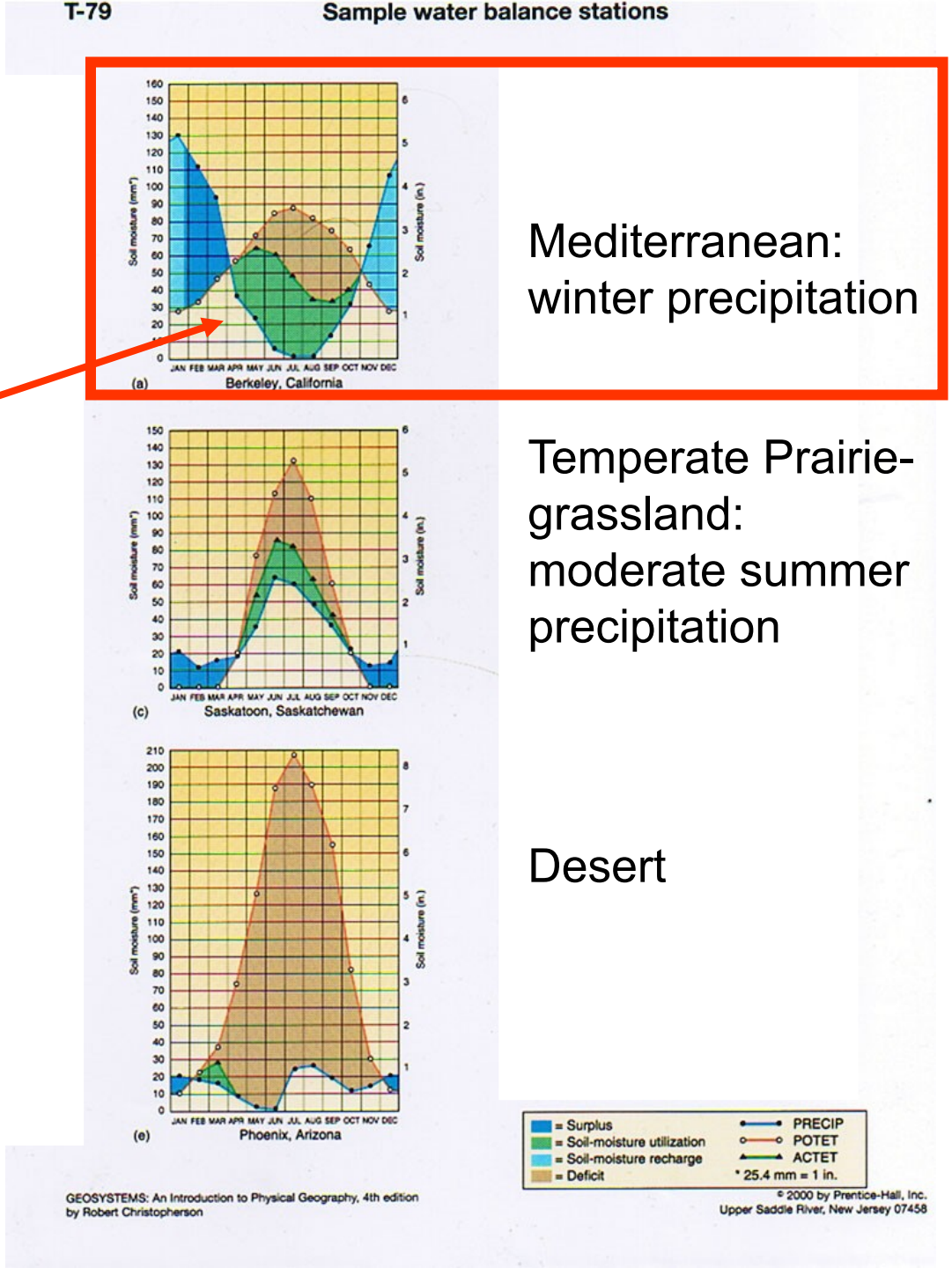
Arid desert of **Phoenix, Arizona**: Most of the year shows water deficit conditions in soil (in brown) ; winter soil moisture stores are quickly used up (green zone) in Feb-Mar and there is a very short period of recharge to soil (Dec –Jan, blue zone).

Ottawa, Ontario (much rainier and cooler climate than Phoenix, AZ):

There is a significant period of recharge (in dark blue) when it rains and it's cold (low Et). A very small Soil water deficit (brown zone) only develops near the end of summer (July August) on average



In brown: soil moisture deficit



Modelled Changes due to Global warming, 30-50 years from now

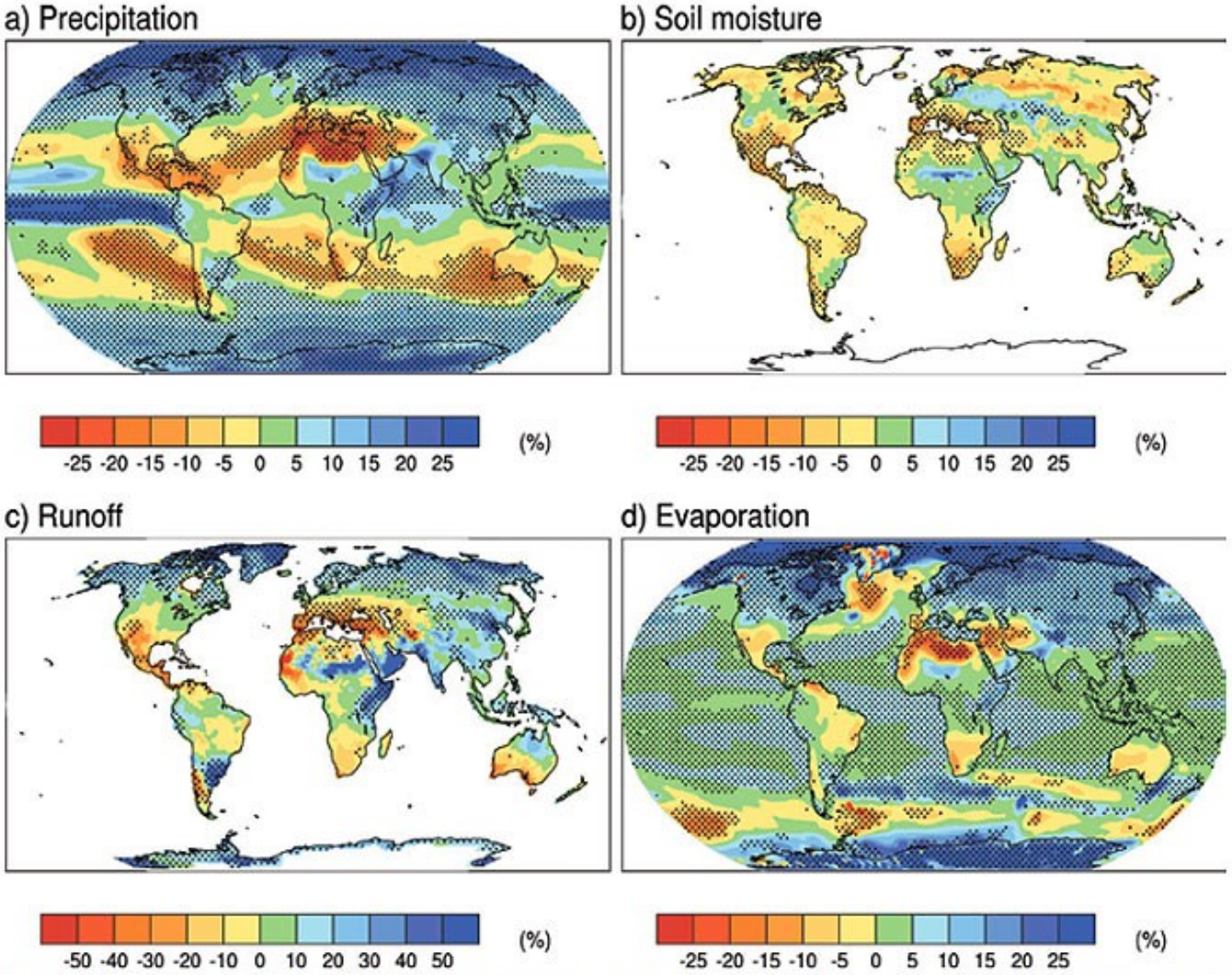
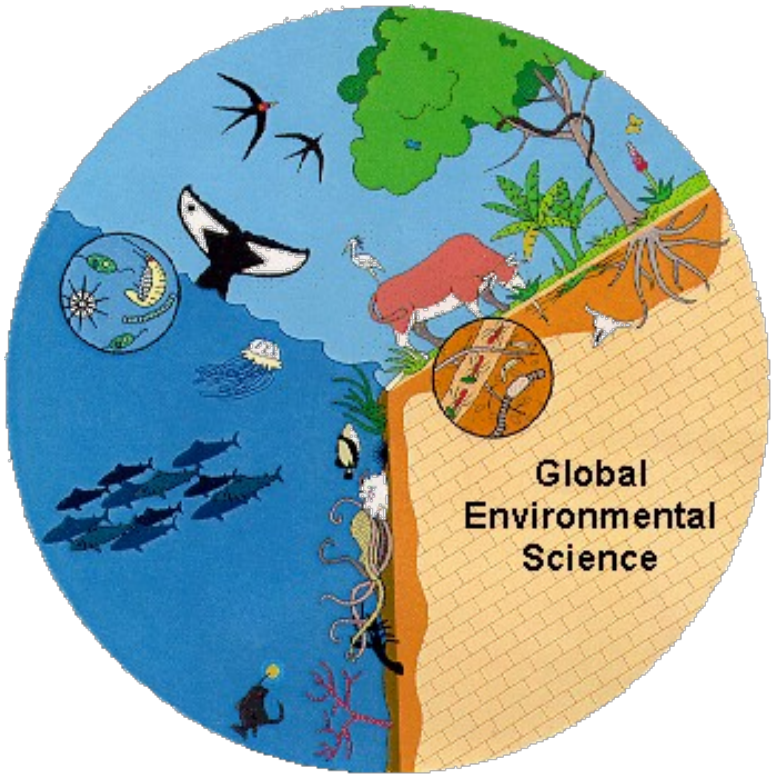


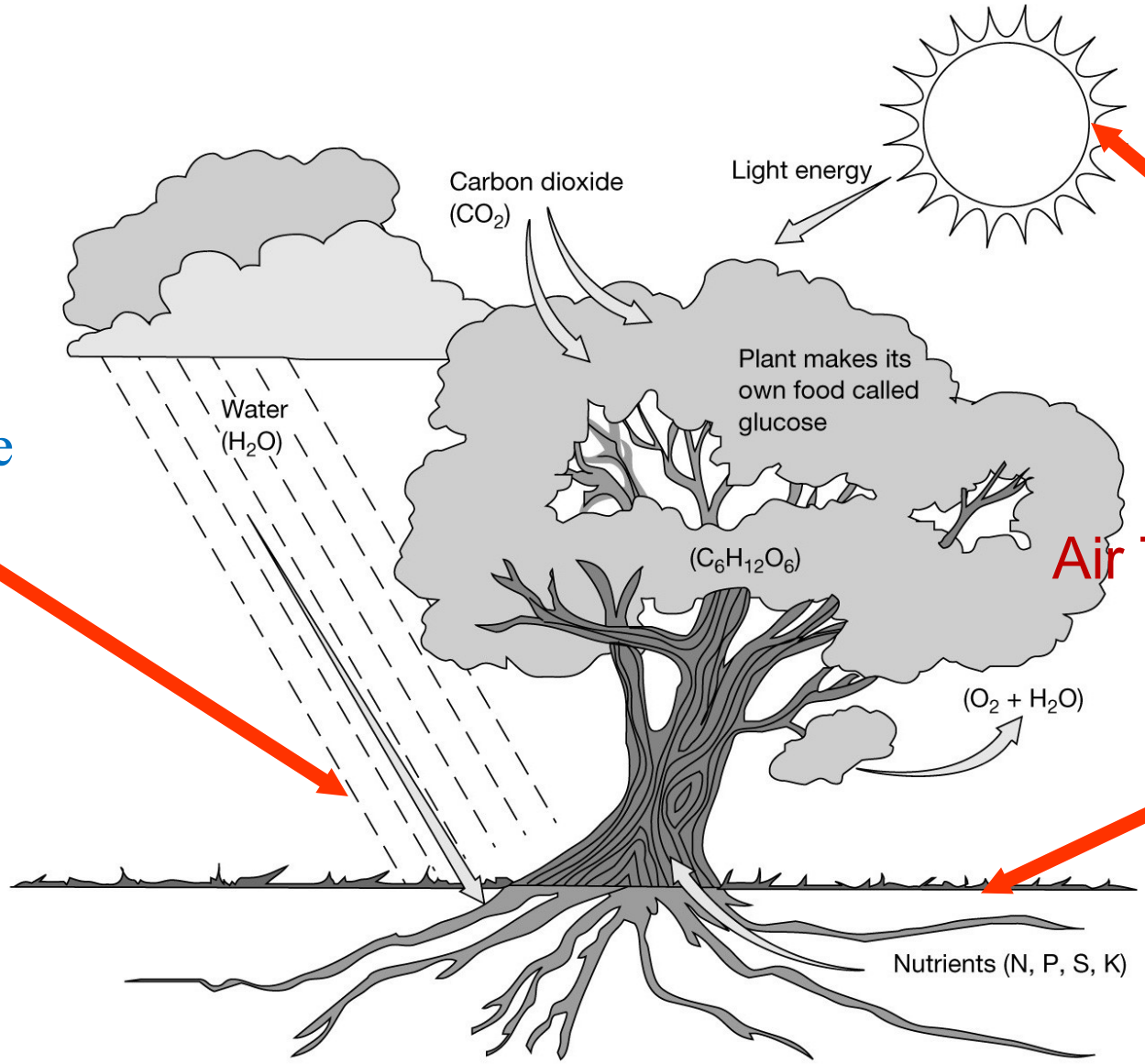
Figure 2.8: Fifteen-model mean changes in (a) precipitation (%), (b) soil moisture content (%), (c) runoff (%), and (d) evaporation (%). To indicate consistency of sign of change, regions are stippled where at least 80% of models agree on the sign of the mean change. Changes are annual means for the scenario SRES A1B for the period 2080–2099 relative to 1980–1999. Soil moisture and runoff changes are shown at land points with valid data from at least ten models. [Based on WGI Figure 10.12]

THE BIOSPHERE
Biomes
Ecosystems

Portion of planet with biota: living organisms
Includes portions of hydro-, atmo-, and litho-spheres



Beyond soil moisture, plants need light, heat, and nutrients from the soil



Moisture in soil

Solar Energy and Heat

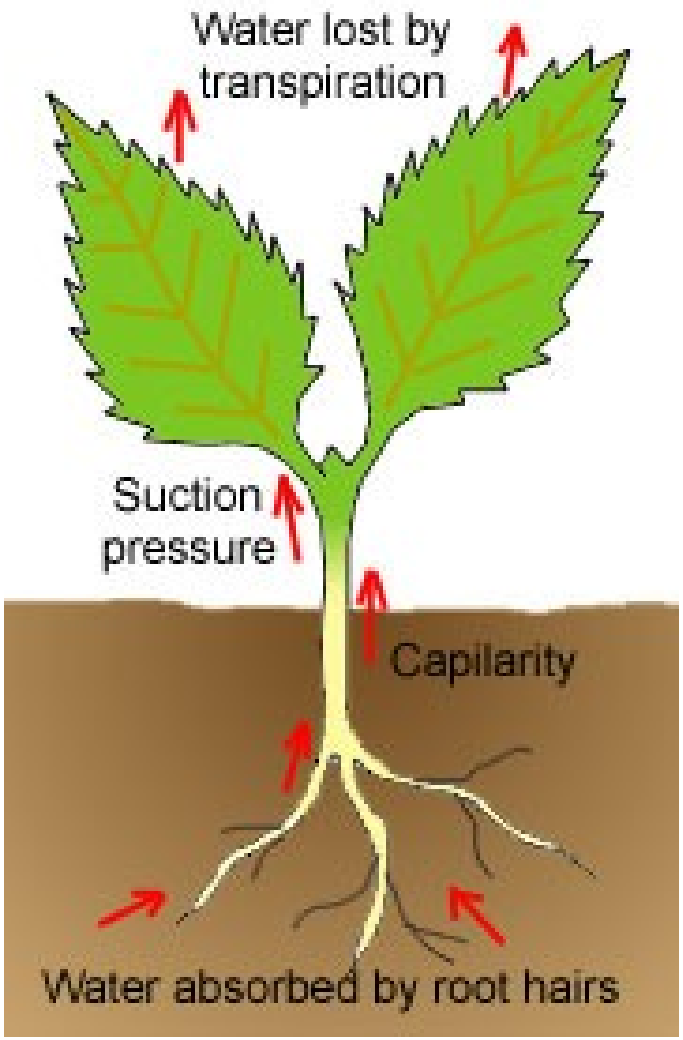
Air Temperature

Chemical Nutrients

Essential plant macro-nutrients

Element	Available from	Source
Carbon C	**Carbon dioxide, CO_2**	**Air or dissolved in water**
Hydrogen H	**Water, H_2O**	**Water**
Oxygen O	**Carbon dioxide, CO_2**	**Air or dissolved in water**
Nitrogen N	**Nitrate NO_3^-** **Ammonium NH_4^+** **N_2 by N fixation**	**Soil minerals or dissolved in H_20** **Air**
Sulfur S	**Sulfate, SO_4^{2-}**	**Soil minerals or dissolved in H_20**
Phosphorus P	**Phosphate PO_4^{3-}**	**Soil minerals or dissolved in H_20**

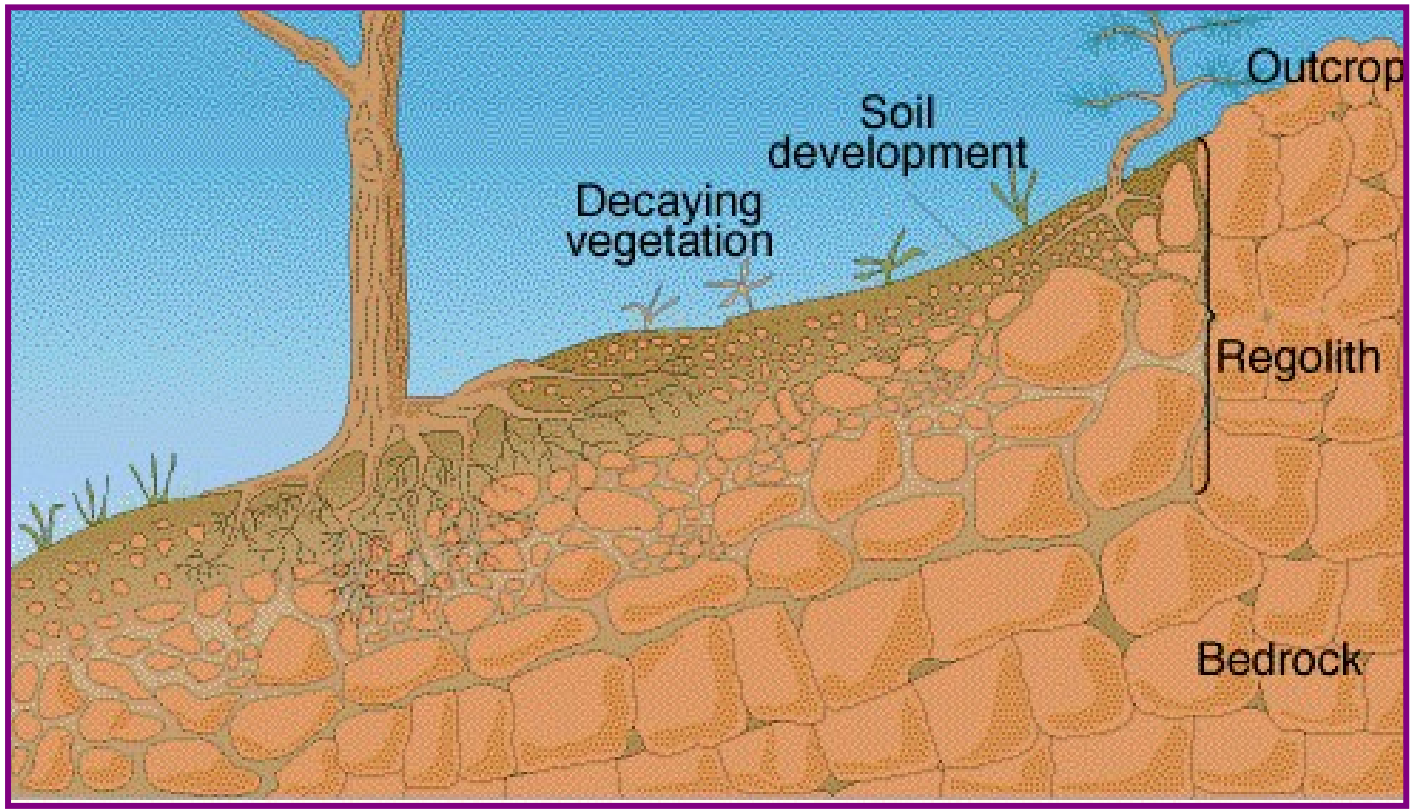
Also many micro-nutrients: Potassium (K), Calcium (Ca) , etc.



Soil water taken up by plant roots as plant transpires brings up to leaves the chemical **nutrients** dissolved in the soil water . In leaves, nutrients are incorporated into the sugars and proteins created through photosynthesis (along with H_2O and CO_2)

Weathering and Soil development

The end product of rock weathering is a physically and chemically altered rock layer called Regolith which overlies and shields the unaltered bedrock underneath.



Source: R.W. Christophersen, 2000 Geosystems

Further weathering and biological activity lead to soil formation.

Chemical Weathering: Hydrolysis

Weak acids strip cations out of minerals

CO_2 + H_2O → H_2CO_3

Carbon dioxide + water → carbonic acid

$2KAlSi_3O_8$ + $2H_2CO_3$ + H_2O →

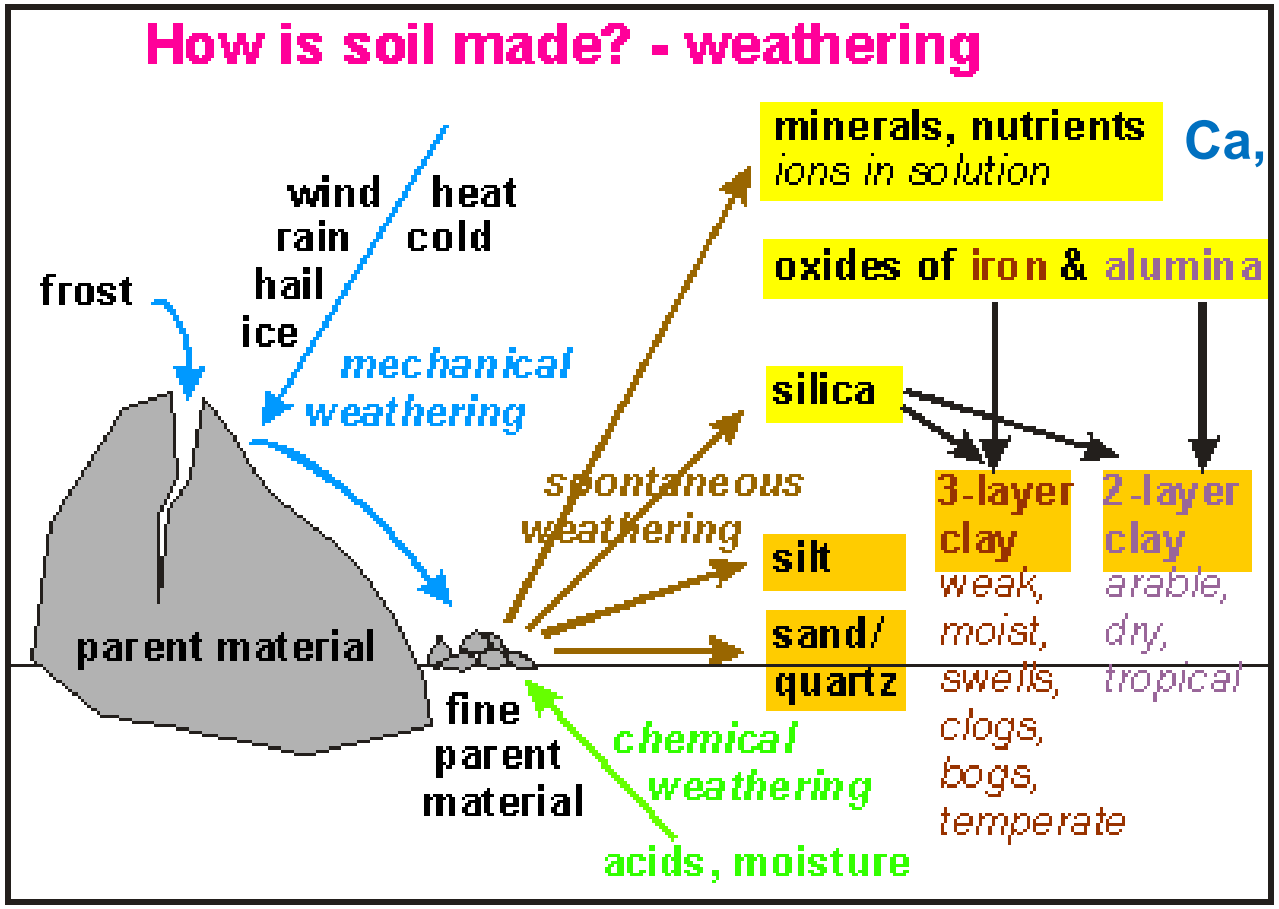
feldspar + carbonic acid + water →

$Al_2Si_2O_5(OH)_4$ + $4SiO_2$ + $2K^+$ + $2\ HCO_3^-$

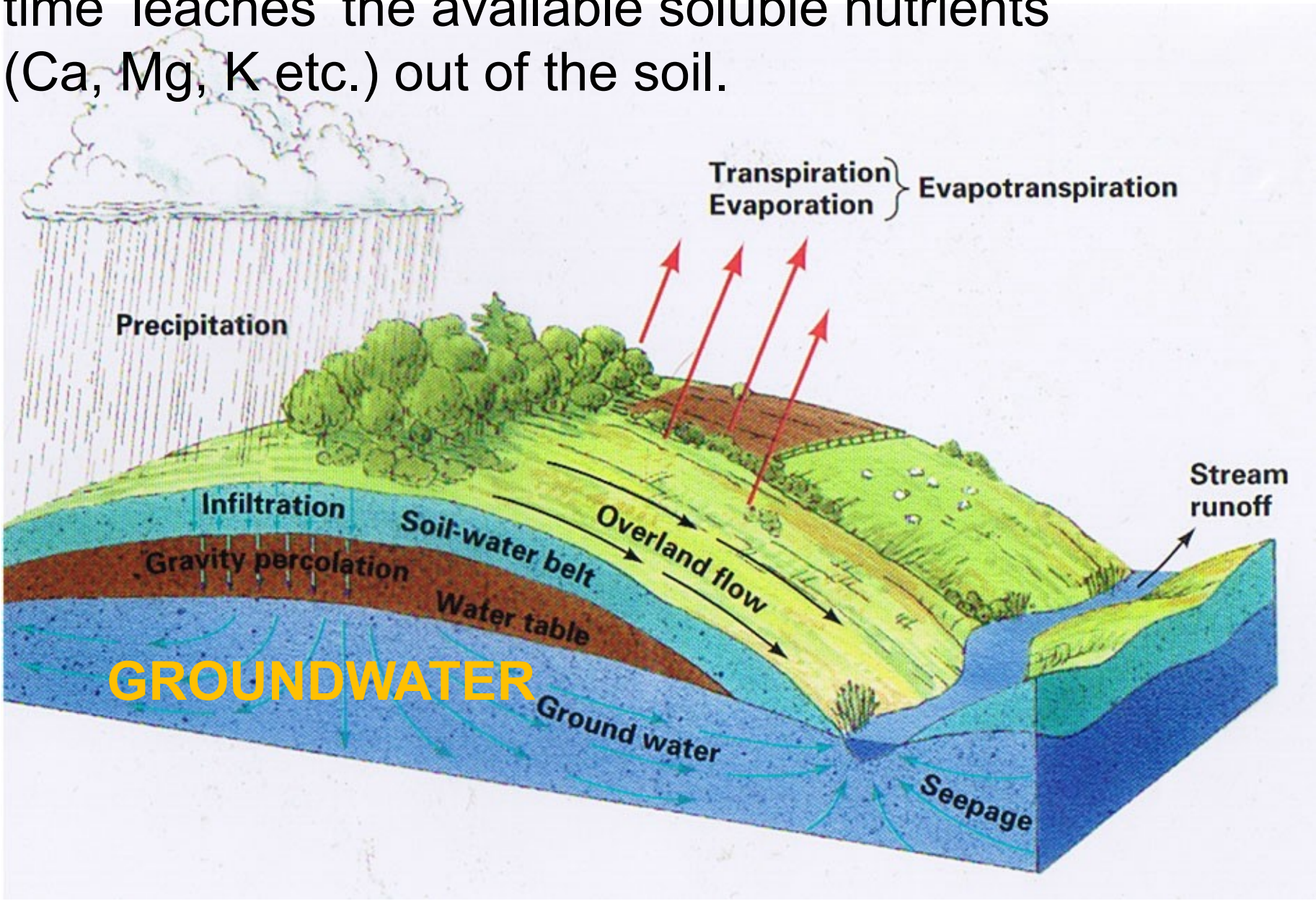
kaolinite (a clay) + silica (dissolved) + potassium (dissolved) + bicarbonate (dissolved)

These reactions requires water to be Present: either as liquid or as vapor

Exposure of rock minerals to ‘weather’ breaks down some unstable minerals

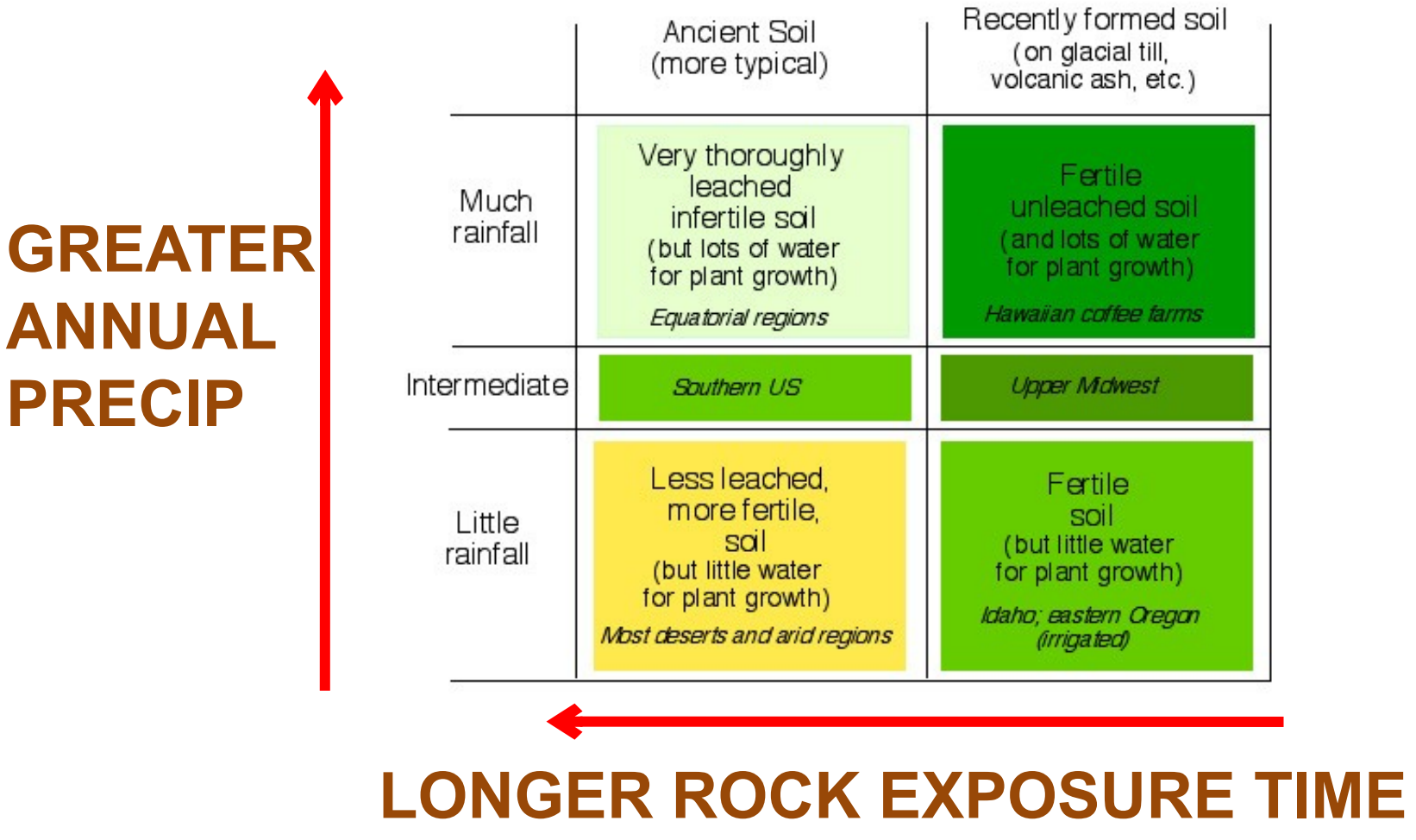


Too much precipitation and groudwater recharge over time 'leaches' the available soluble nutrients (Ca, Mg, K etc.) out of the soil.



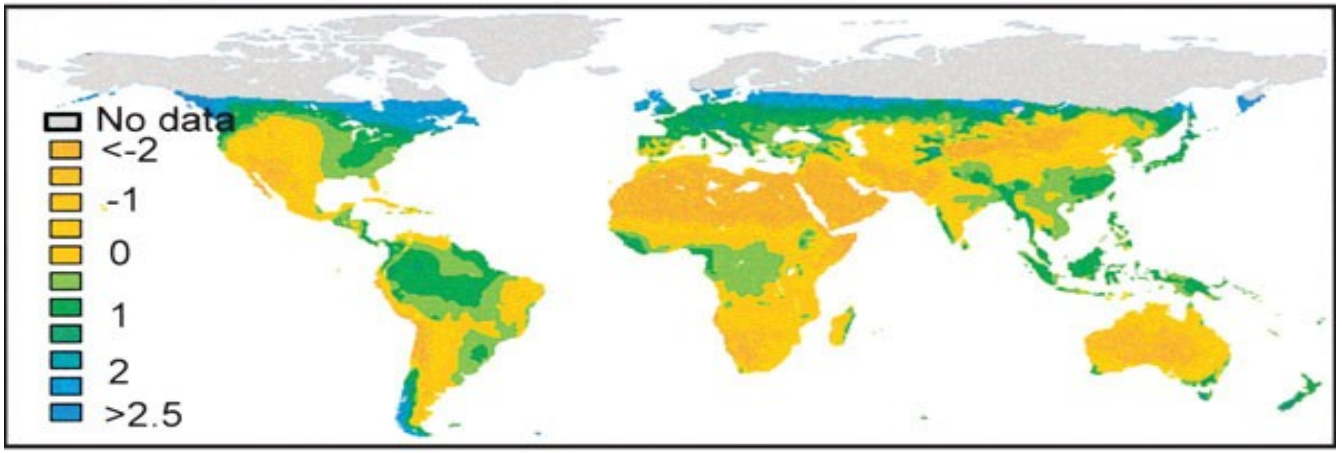
Source: Strahler and Strahler, 2005 Physical Geography, 3rd Edition

The duration of weathering and amount of precipitation affect the inherent soil fertility (available nutrients) (fertility in shades of green, here)



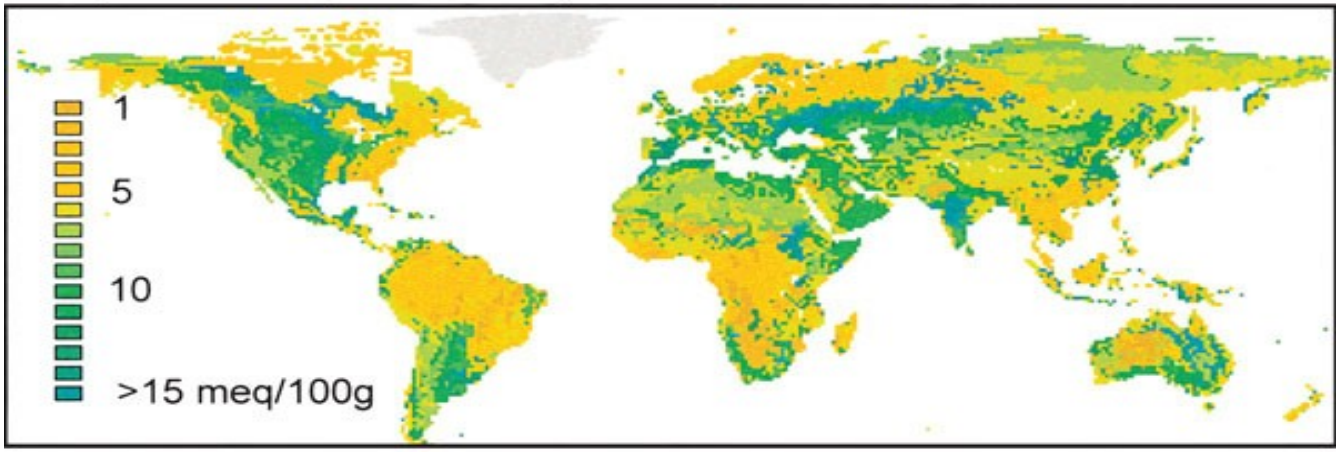
Soil water availability (top map)

a Observed index for plant-available moisture
(log [rainfall/potential evapotranspiration])

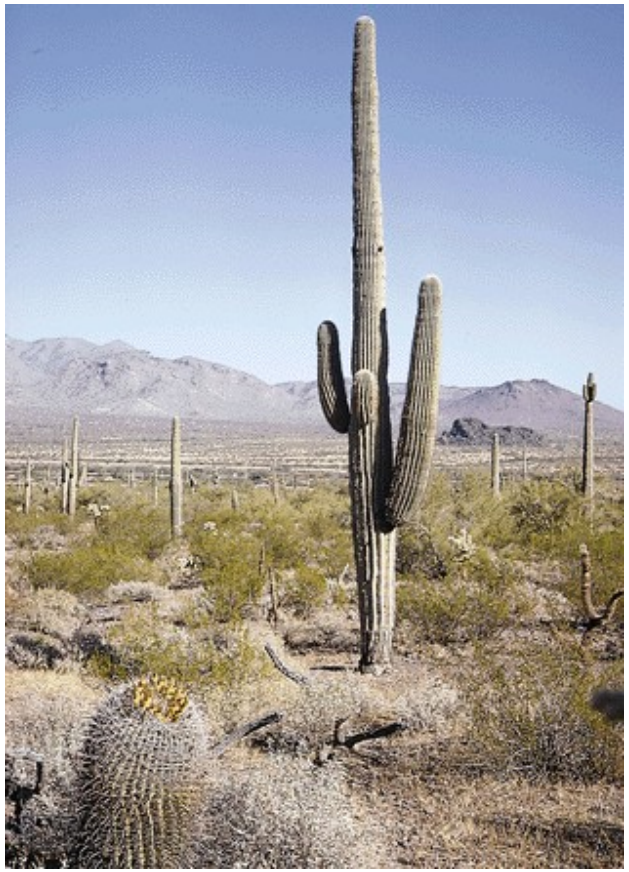


Global patterns in natural soil fertility (below)

b Observed index for plant-available nutrients
(total exchangeable bases)



Biomes



Desert



Grassland

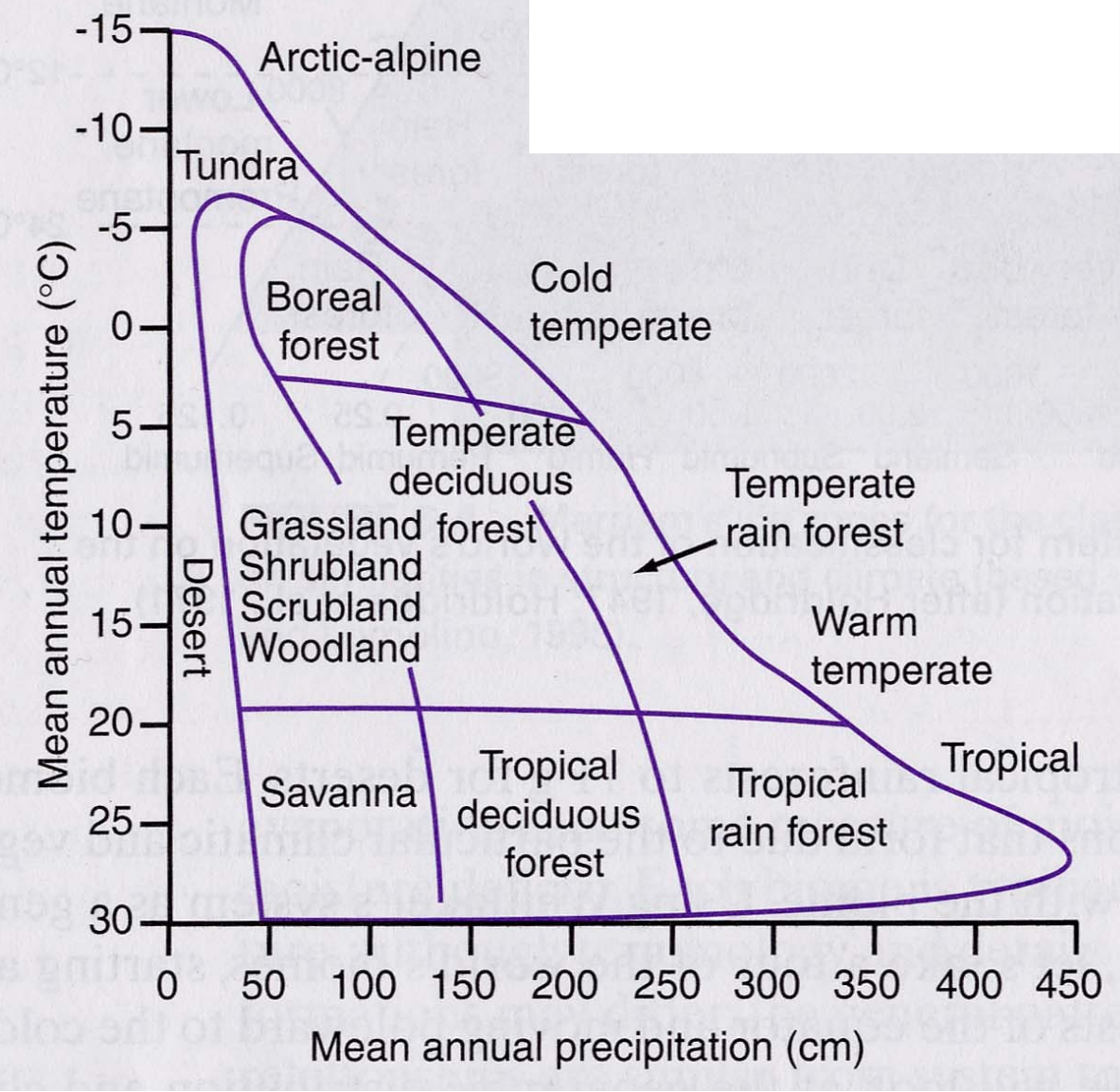
Biome

BIOME = “A large terrestrial ecosystem characterized by **specific plant communities and formations**; usually named after the predominant vegetation in the region.”

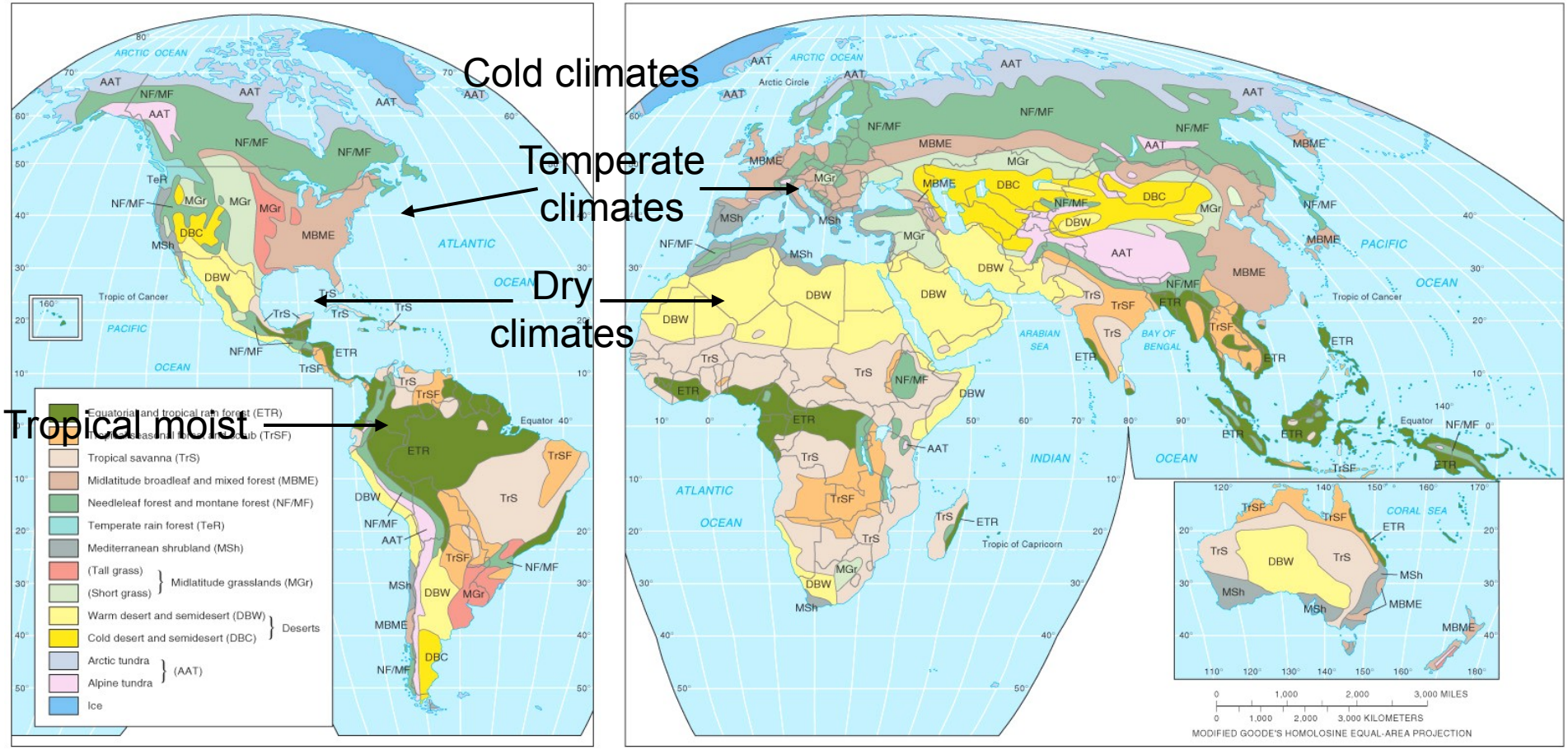
Usually this is associated with **characteristic animal fauna** able to exploit typical habitats and foods in these biomes

Biome: the largest ecosystem grouping

A large terrestrial ecosystem characterized by specific plant communities; usually named after the predominant vegetation in the region.”



Major Terrestrial Biomes



The boreal forest biome





The temperate forest biome

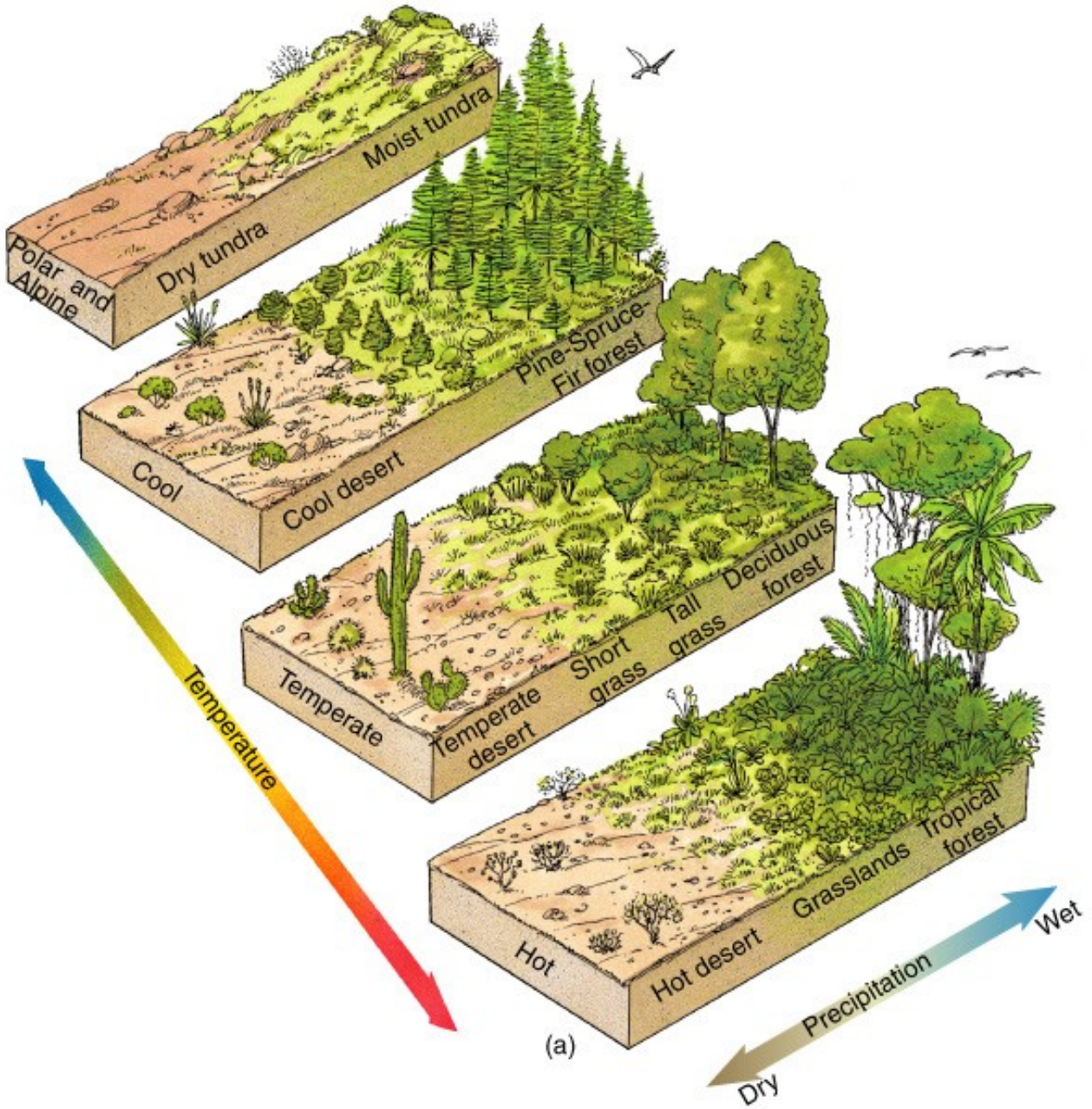
Arctic Tundra





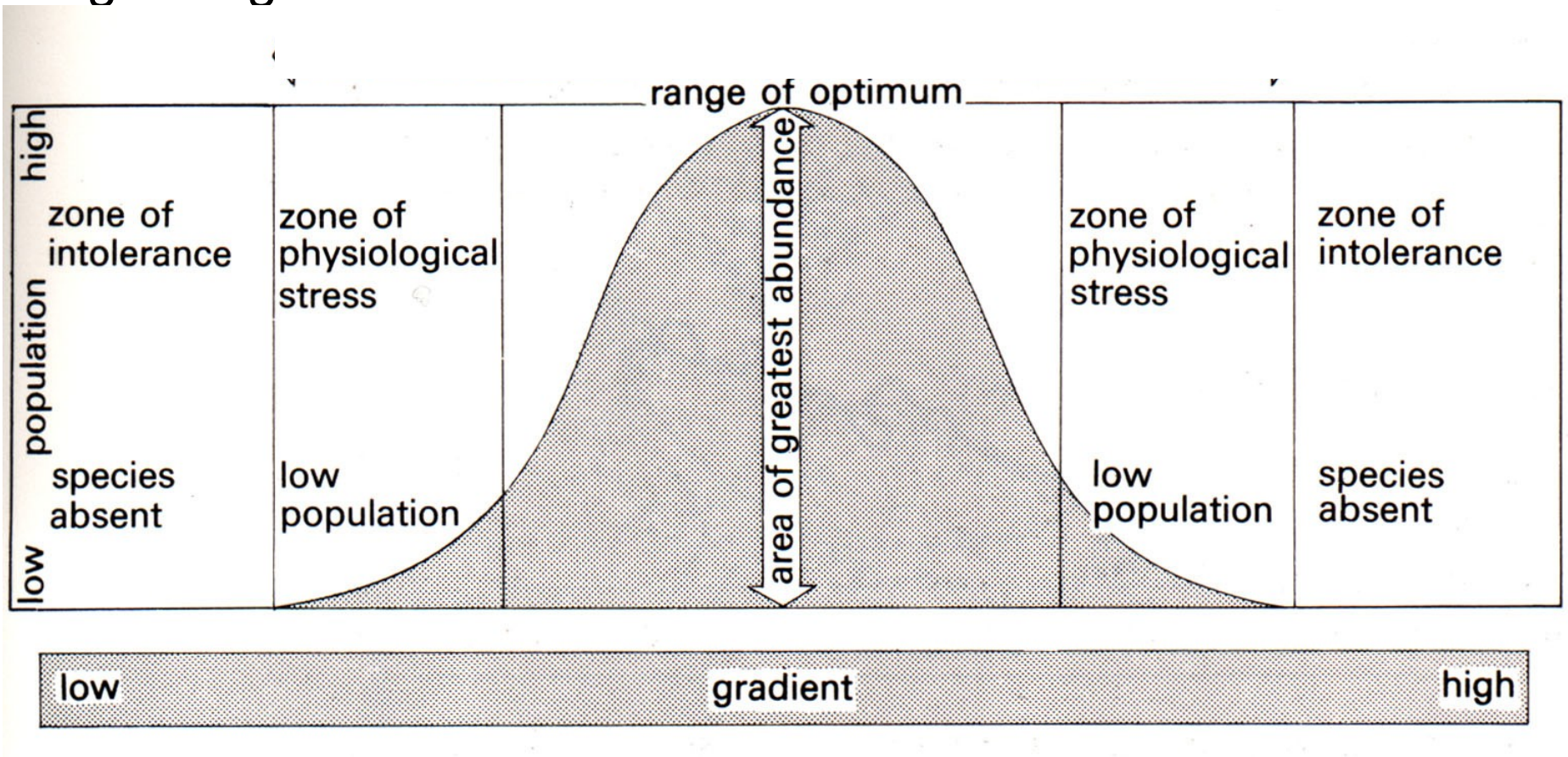
Desert basin, Arizona

Biome response to temperature & precipitation

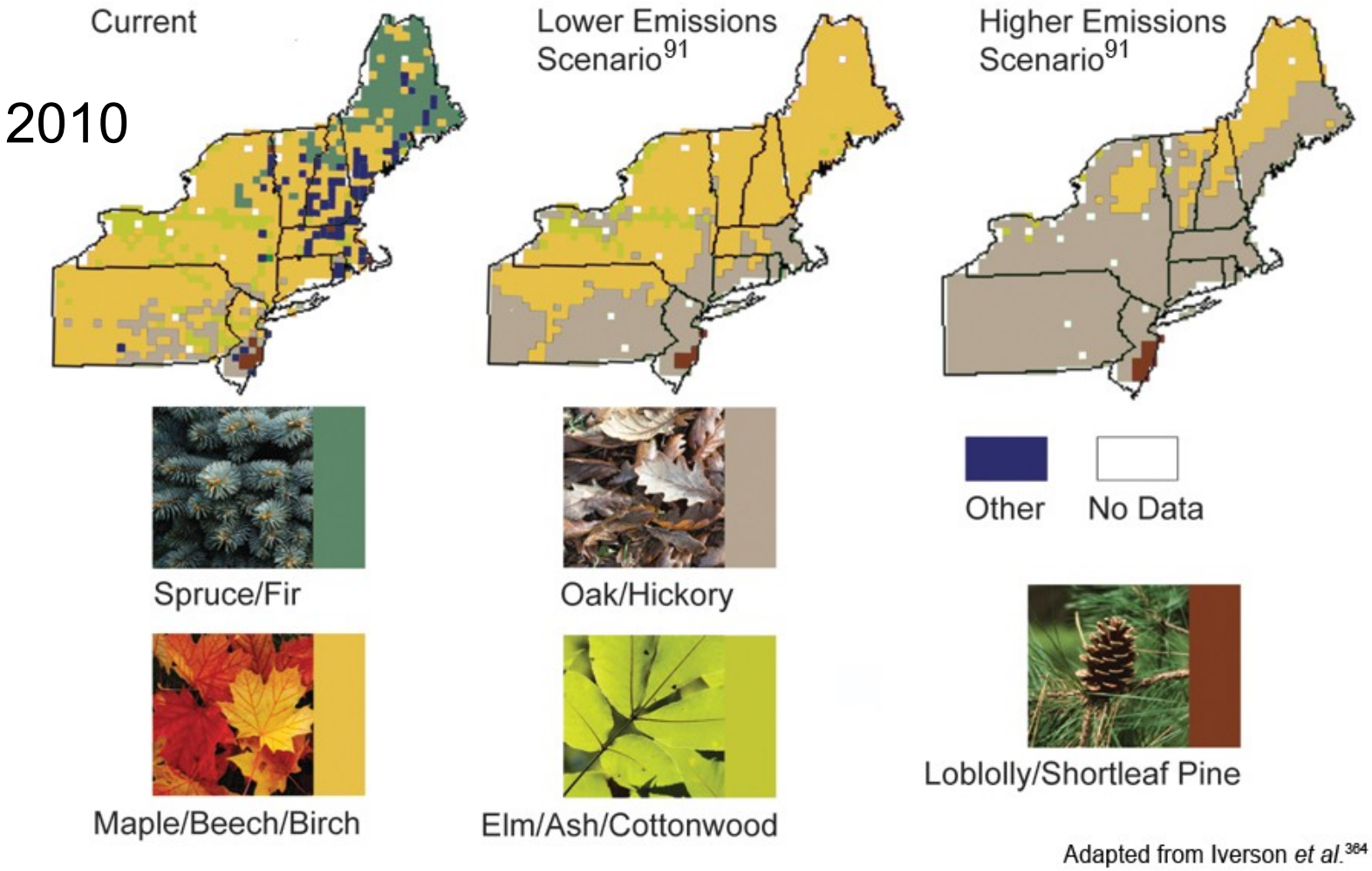


Plants and abiotic factors: temperature, moisture, Etc.

Different species live under a wide variety of conditions. They have different tolerances, and different optimum growing conditions.



2050: Optimal Forest Ranges



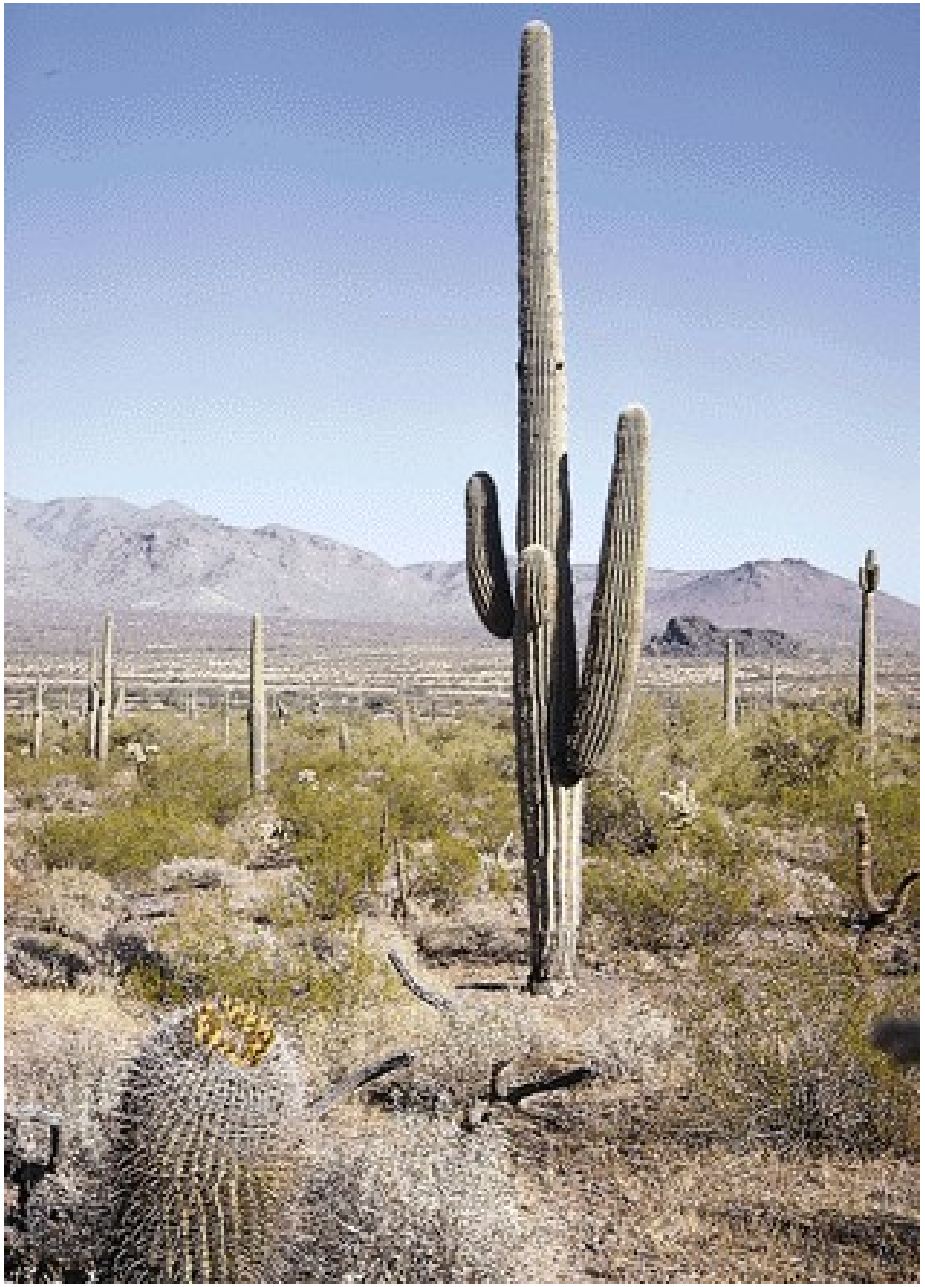
Adapted from Iverson *et al.*[364]

The optimal Sugar Maple range will be 400 kilometers north of today.

Adaptations to dry (or droughty conditions)

To be successful in arid environments, plants must minimize water loss.

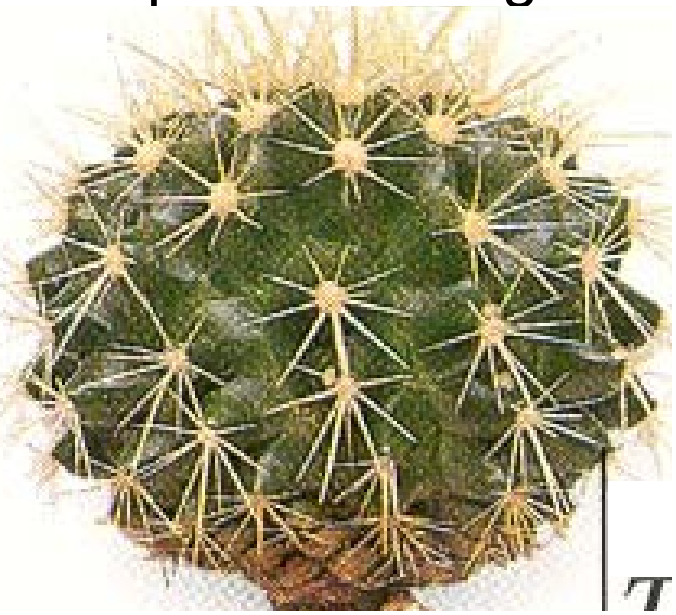
They do this through several mechanisms, or strategies.

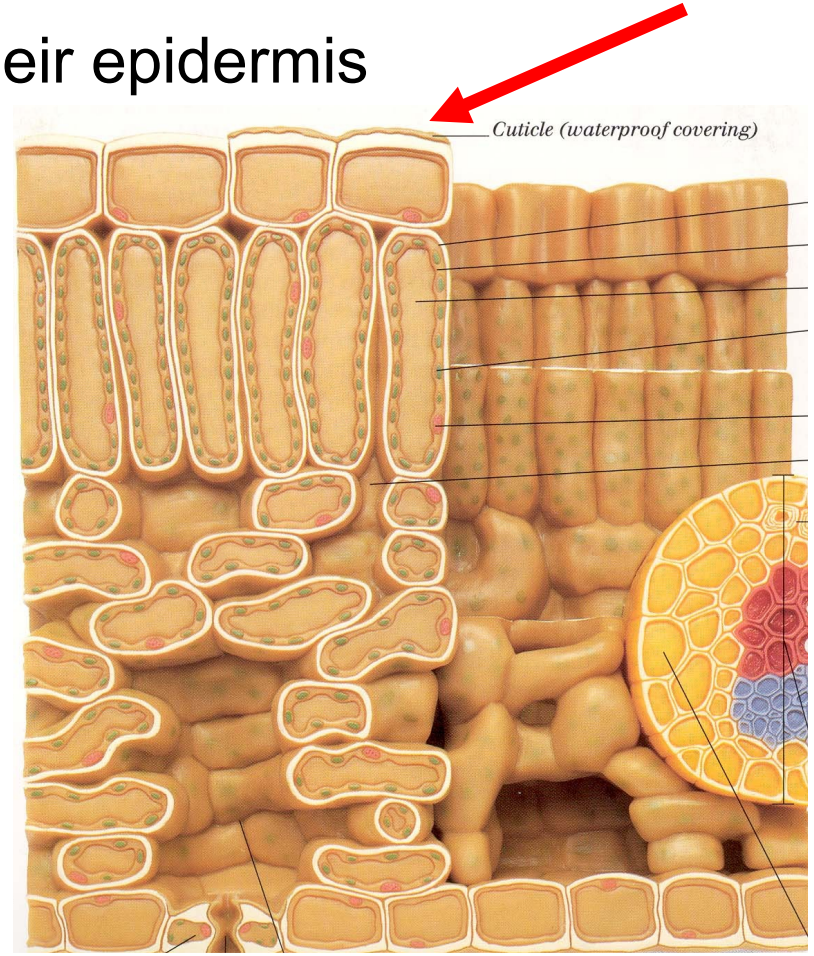


Strategies to reduce water loss

minimize loss of water through their epidermis

- less surface area
- maybe no leaves at all
- thick epidermis
- special coating



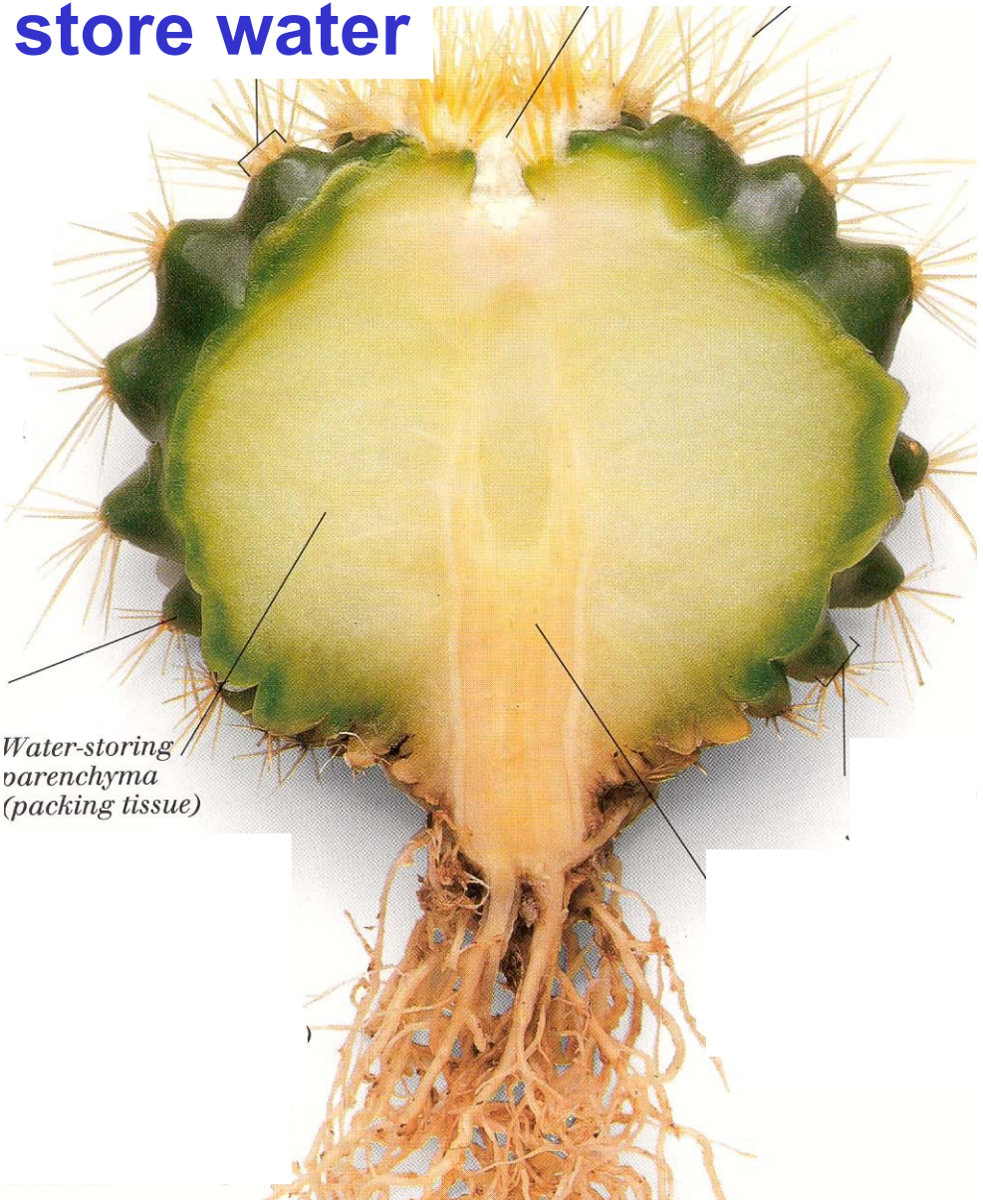


Strategies to obtain and store water

Some desert plants have deep roots that allow them to “tap” groundwater.

Many plants are succulents

They store water in leaves, stems, or roots.



Death Valley: Once in decades bloom after El-Nino related rains, Oct 2015 (again in Fall 2025)







Atacama Desert

Average annual precipitation ~1 cm

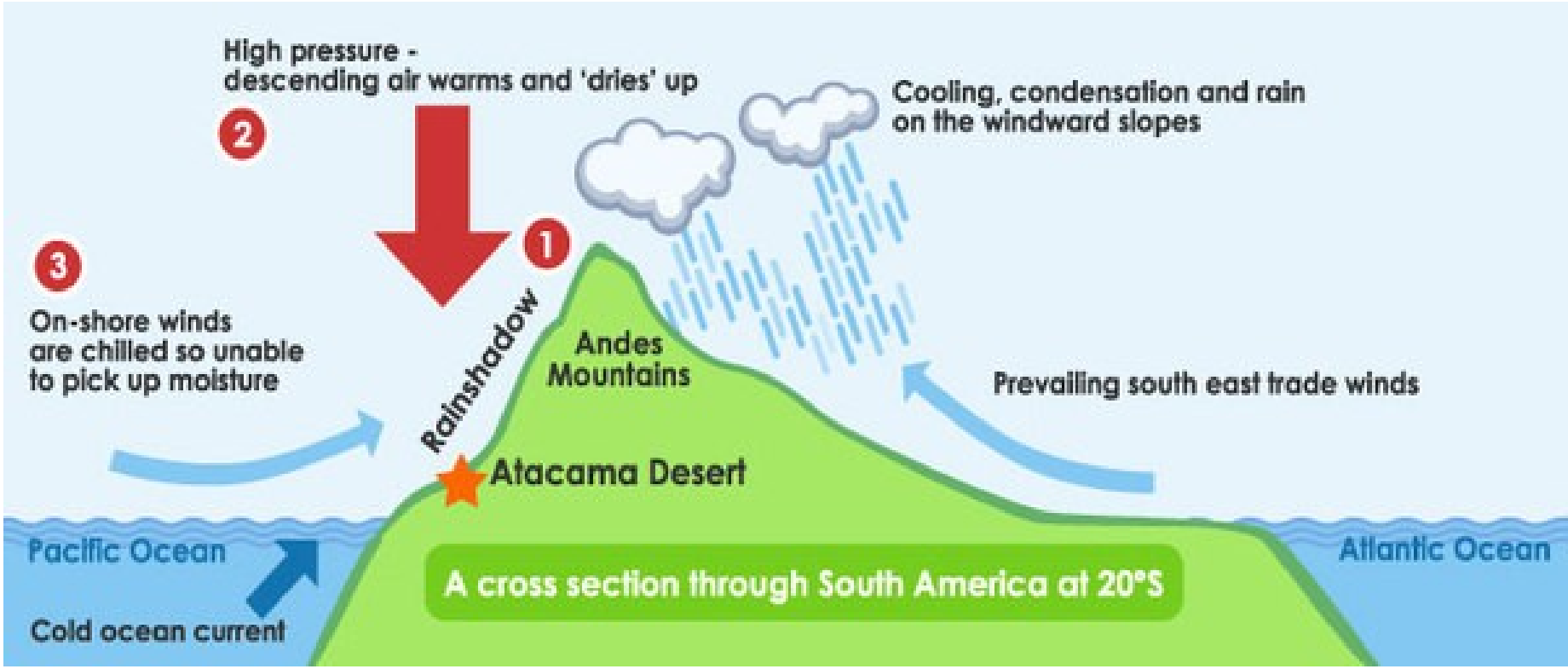




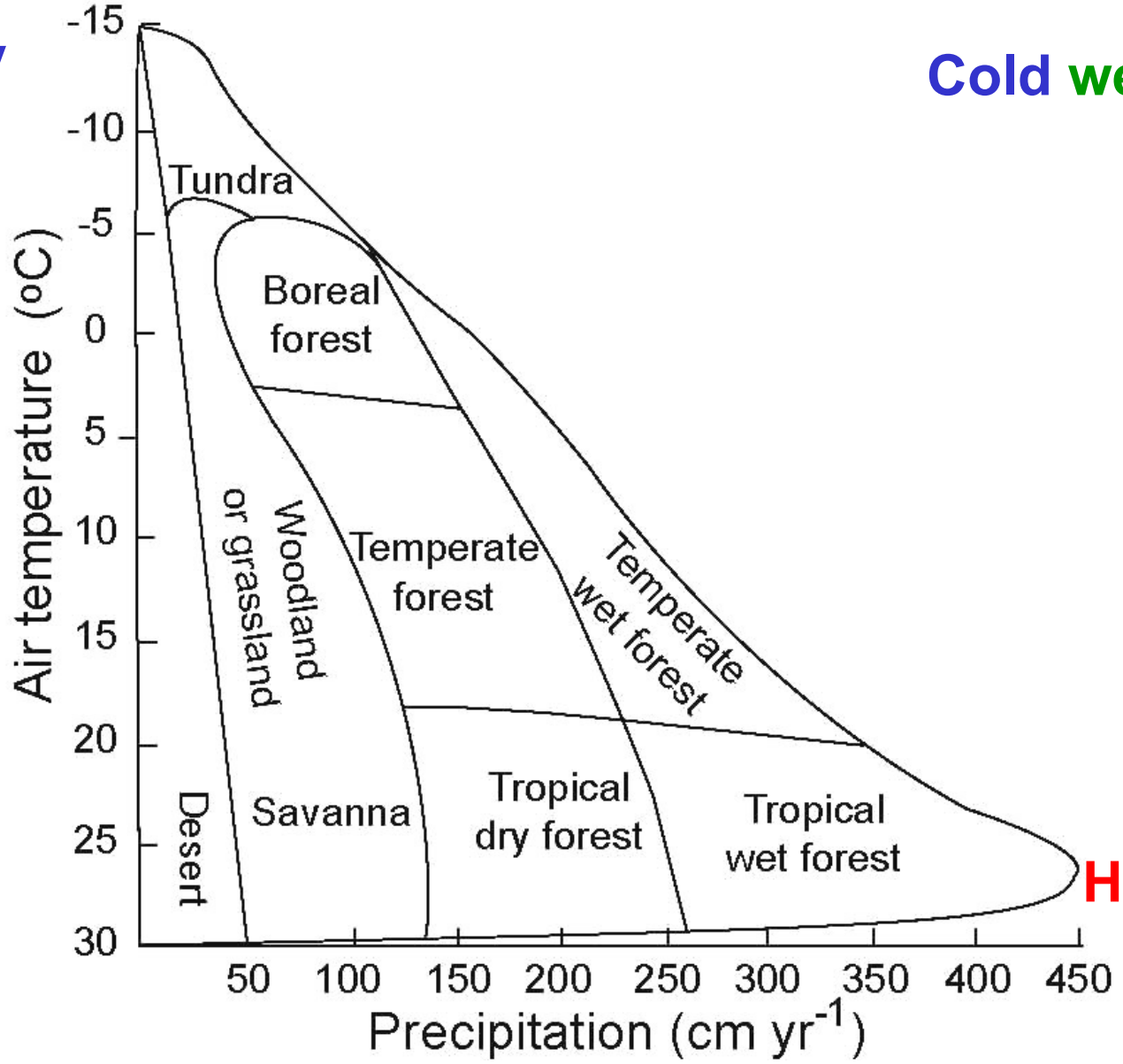


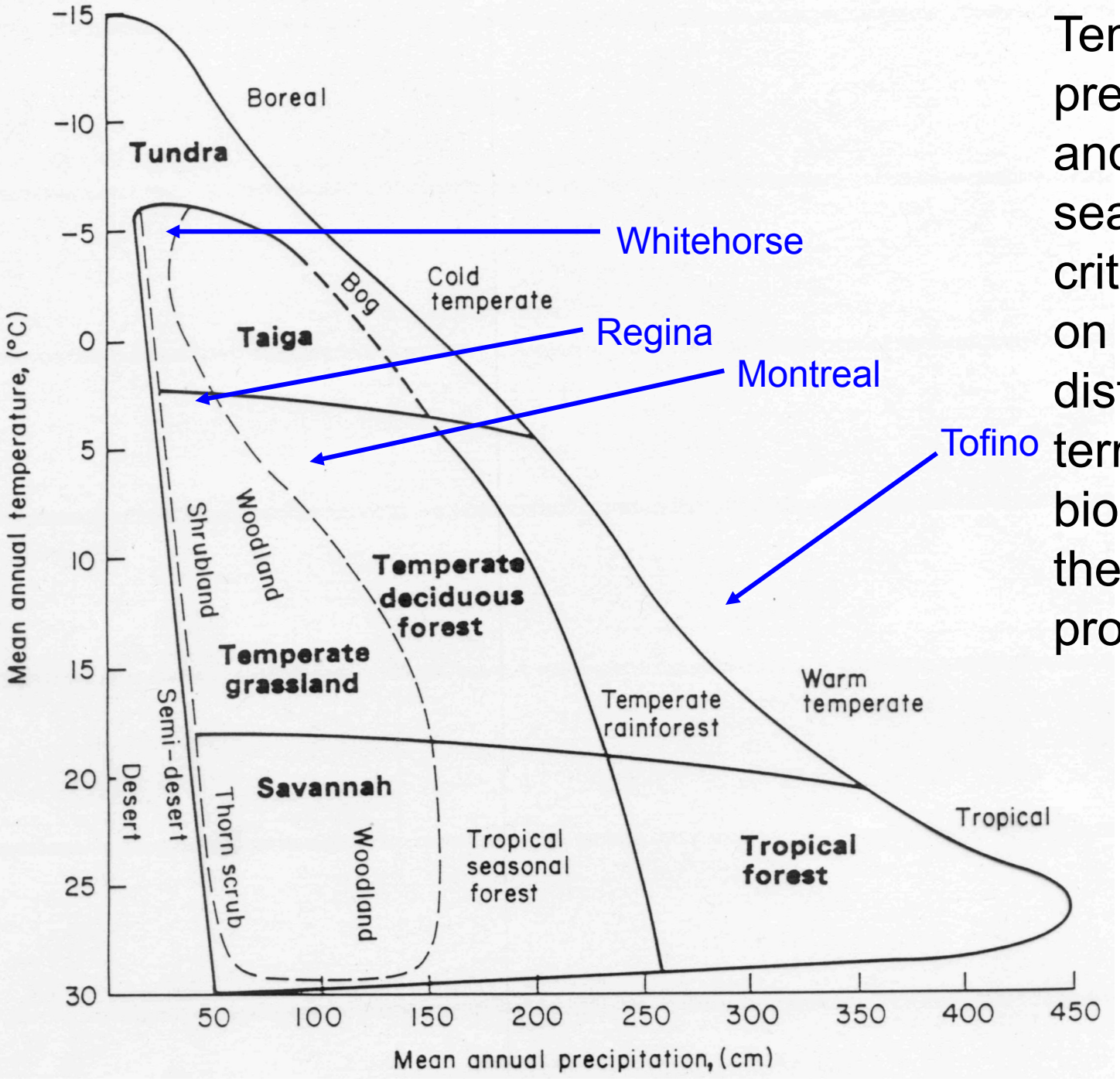
Atacama Desert

20°south:
Suptropical high pressure system
Plus rainshadow effect
Plus cold coastal current



The relationship between biomes & climate

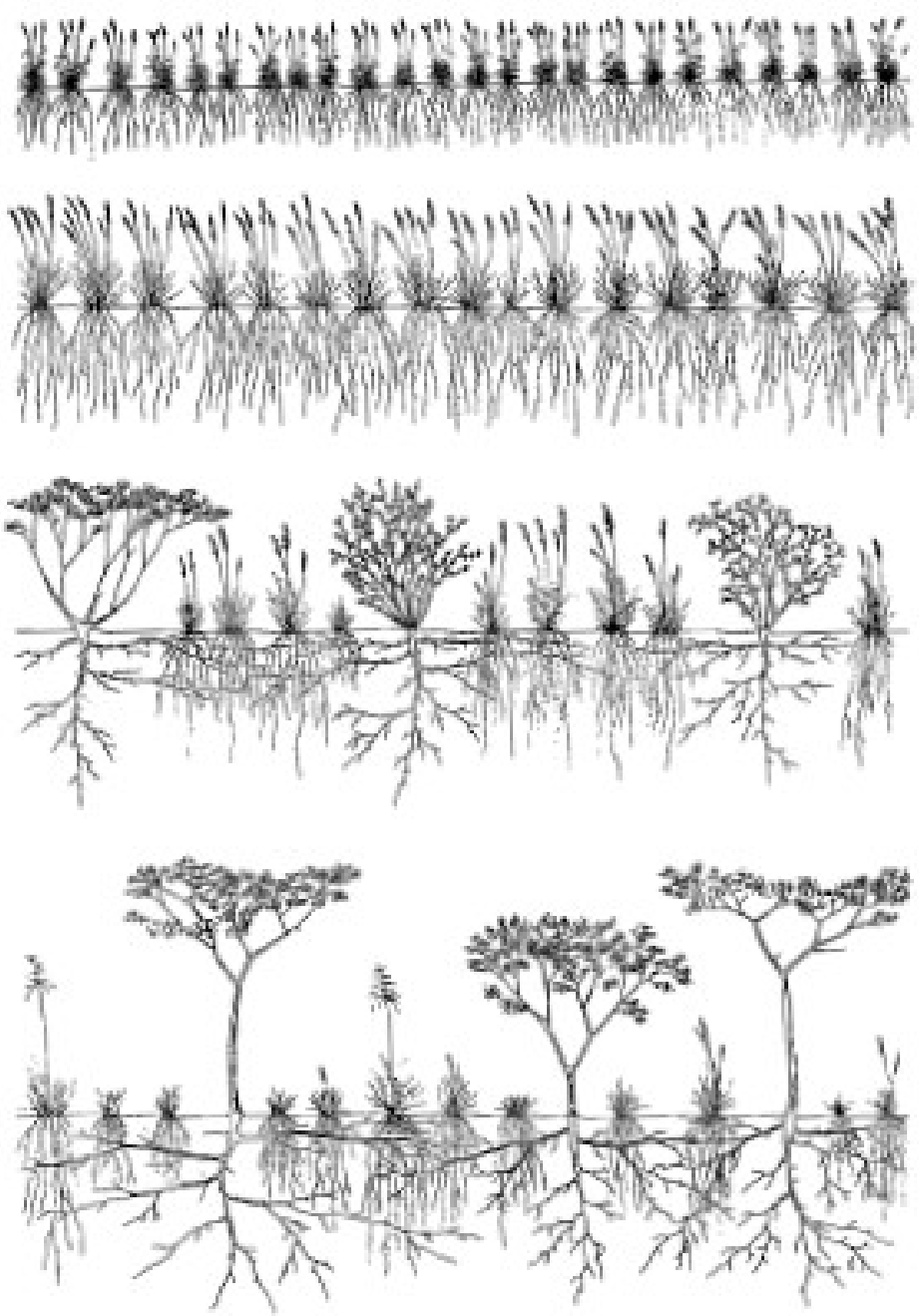




Temperature, precipitation, and their seasonality are critical controls on the distribution of terrestrial biomes and on their productivity.

Competing forms of vegetation cover: grasslands versus forests

Competition for soil water and sunlight



g. 48. Schematic representation of transition from (a and b) grassland to (c) shrub savanna and
) tree savanna. For explanation, see text. (From Walter 1939)

Increasing soil moisture levels helps shift the natural balance from grasses to trees

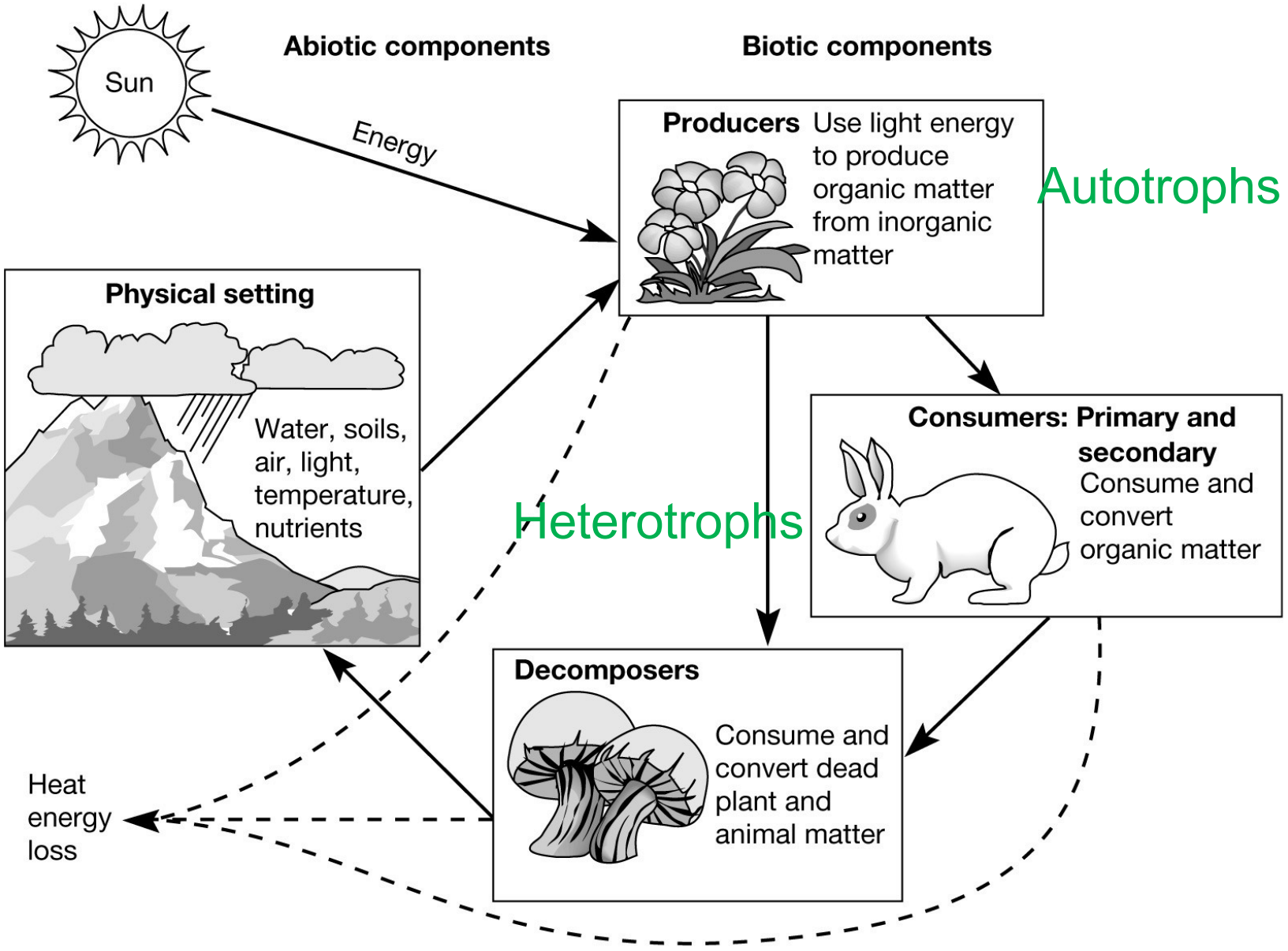
Ecosystems

- A conceptual tool for ecologists studying an area. It's a bounded ecological system consisting of all the organisms in an area and the physical environment with which they interact

- For example: terrestrial ecosystems (a valley for example) involve topography, runoff, soil, atmosphere, climate as well as animals and plants that are adapted to this environment.

- Aquatic ecosytems could be a lake and its stream inflows and outflows; or a given marsh, etc.

Ecosystem Components



Heterotrophs









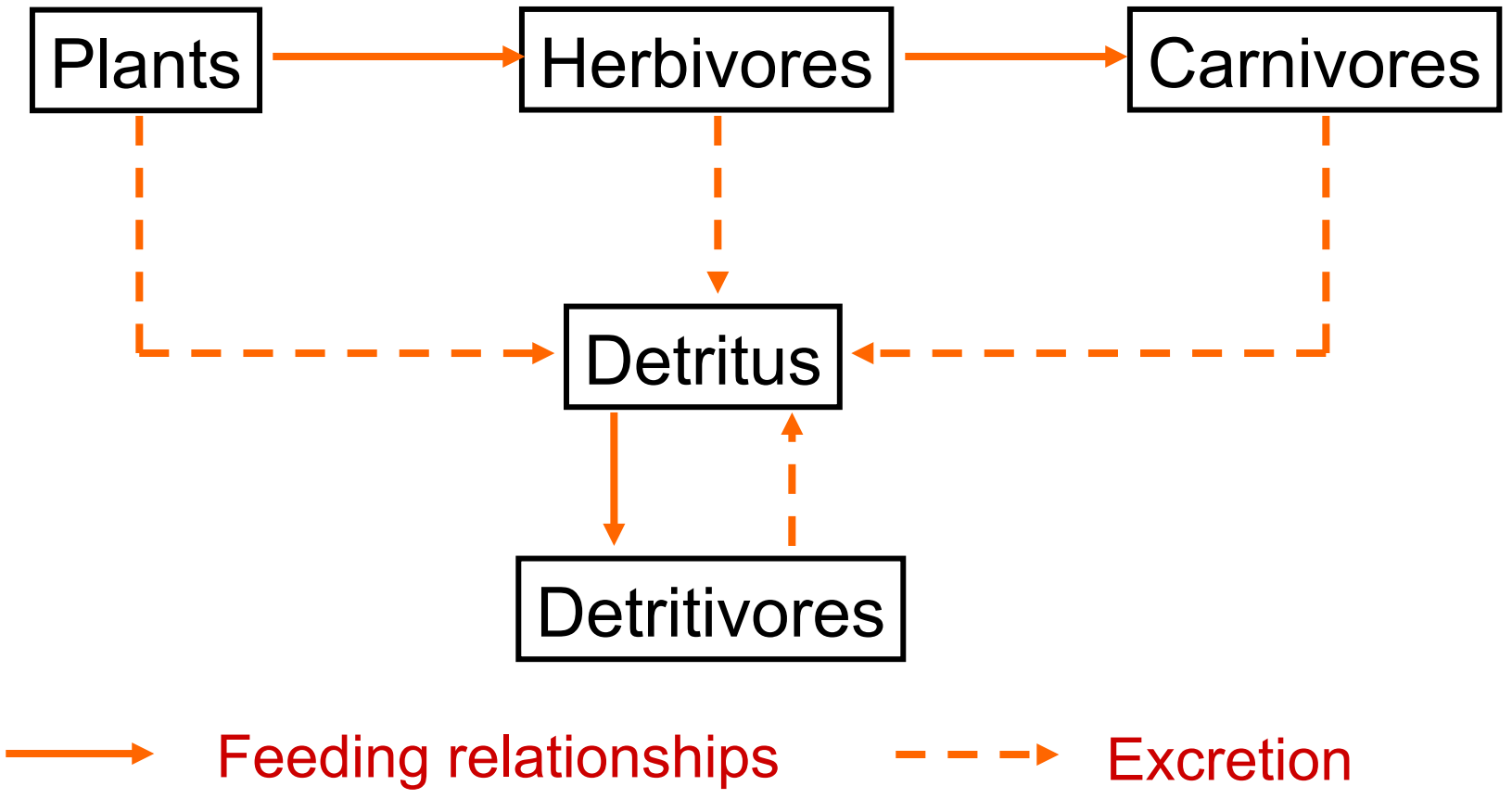
Plant grazers are ‘heterotrophs’, as are their predators

Abiotic and Biotic Components of Ecosystems

ABIOTIC COMPONENTS	BIOTIC COMPONENTS
Sunlight	Plants (Primary producers)
Temperature	Herbivores (Primary Consumers)
Precipitation	Carnivores
Water or moisture	Omnivores (Secondary Consumers)
Soil or water chemistry (e.g., NH_4+)	Detritivores (Decomposers)
Etc.	

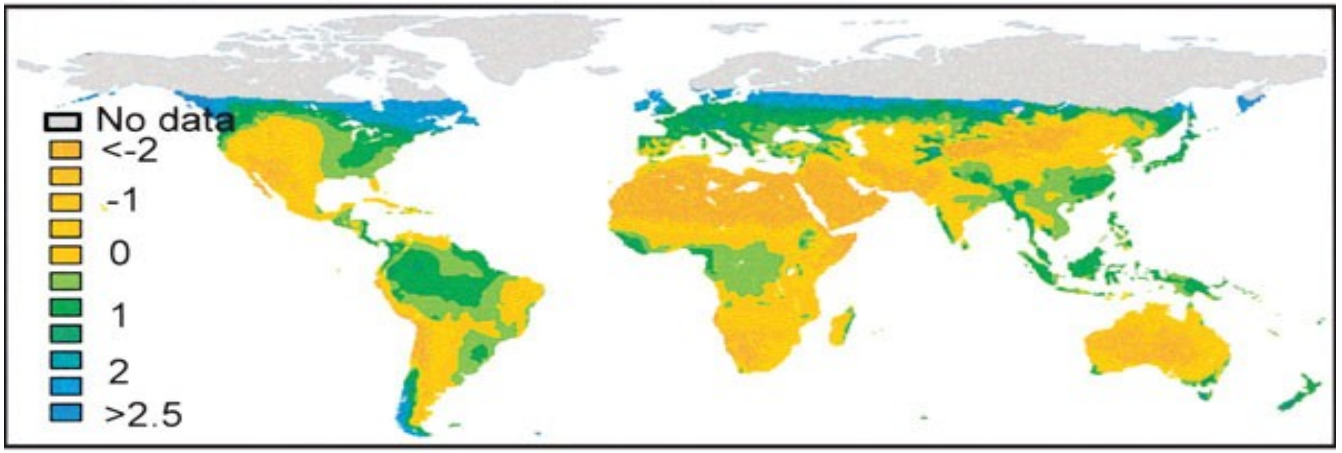
All of these vary over space/time but are 'balanced'; they are in long term steady state

Interactions of Detrital and Grazing (herbivore) Food Chains



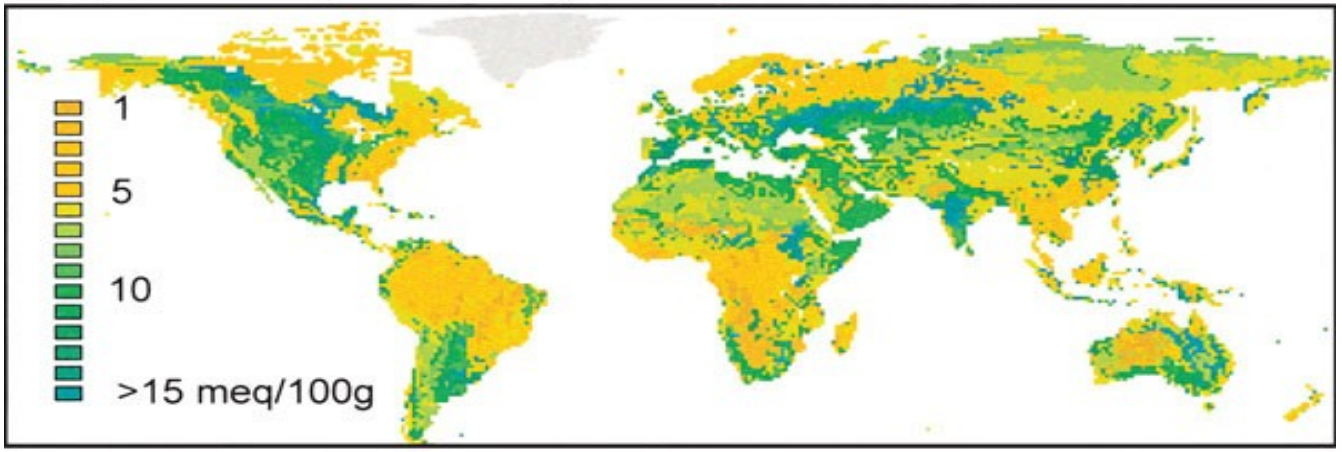
Soil water availability (top map)

a Observed index for plant-available moisture
(log [rainfall/potential evapotranspiration])



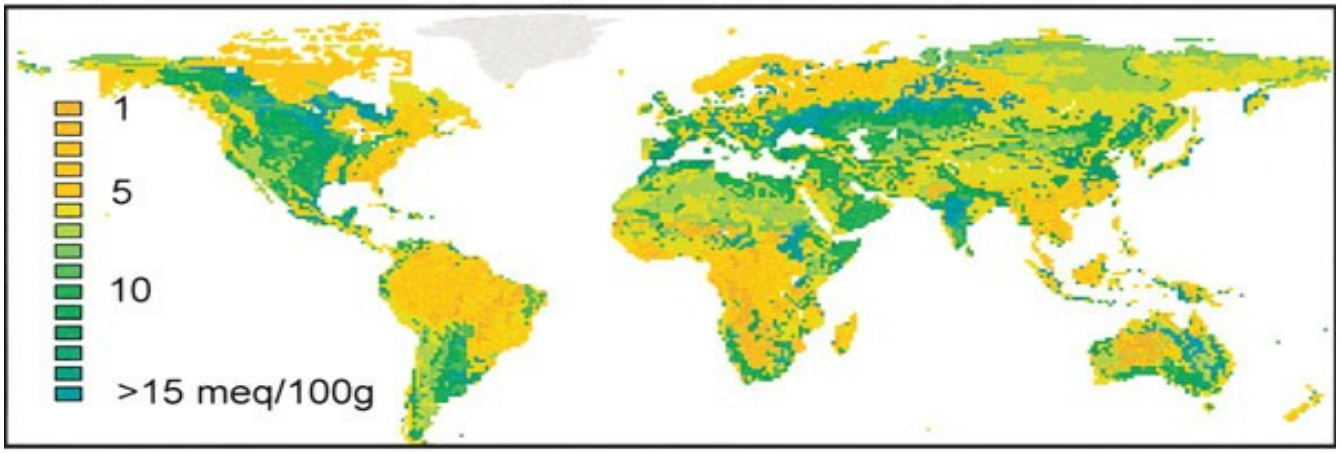
Global patterns in natural soil fertility (below)

b Observed index for plant-available nutrients
(total exchangeable bases)



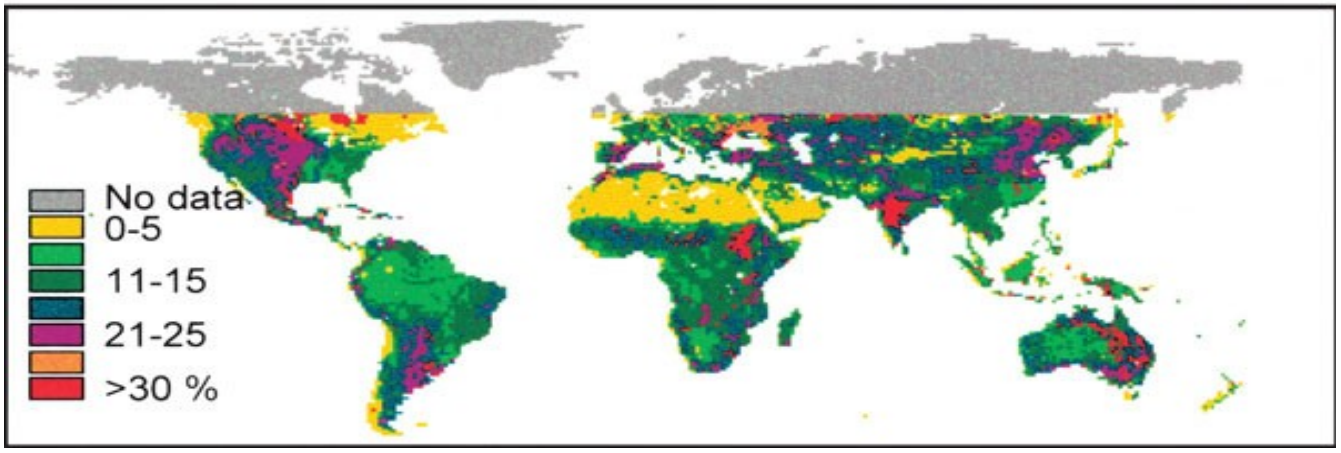
Global patterns in natural soil fertility (below)

b Observed index for plant-available nutrients
(total exchangeable bases)

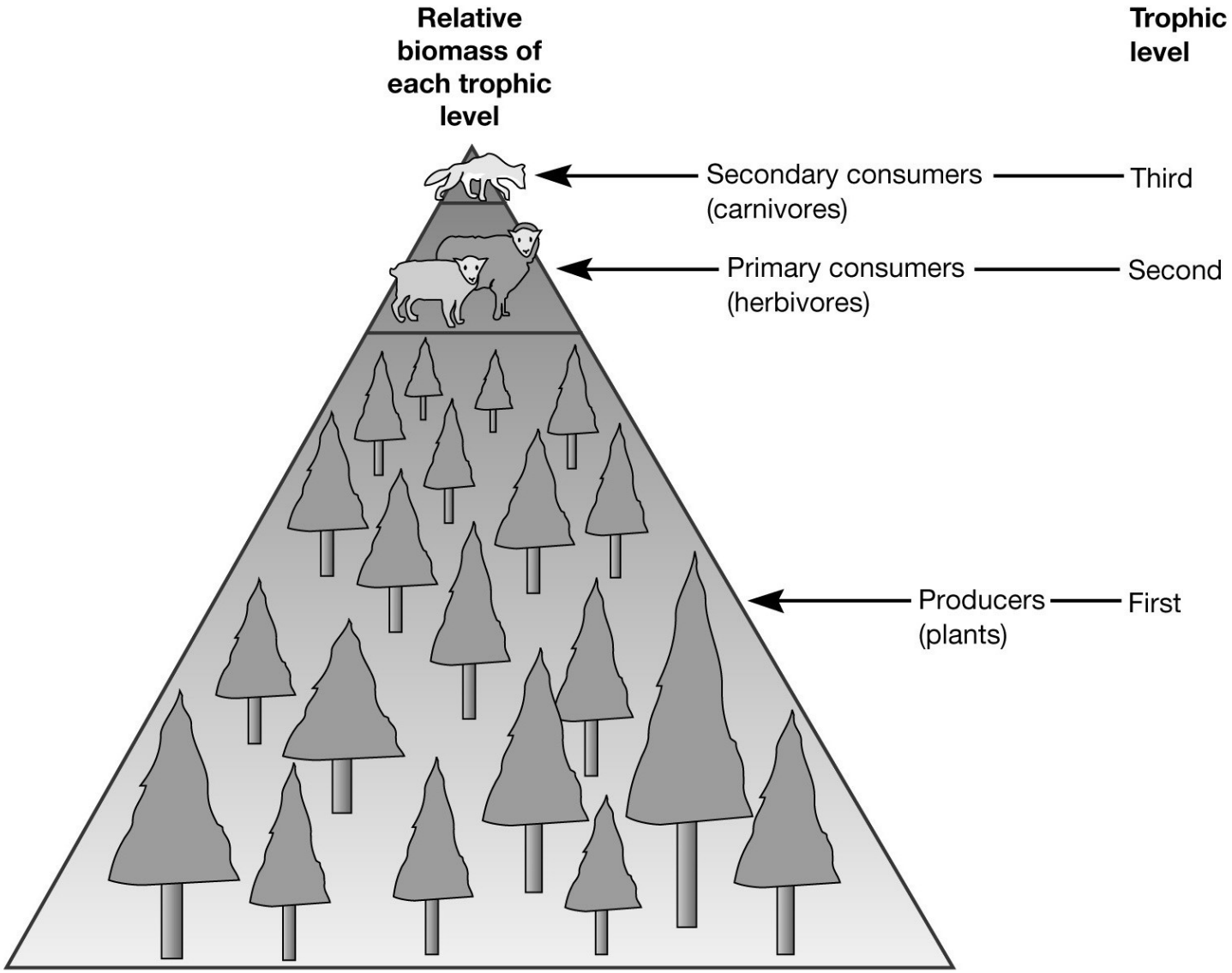


Proportion of species that are large herbivores

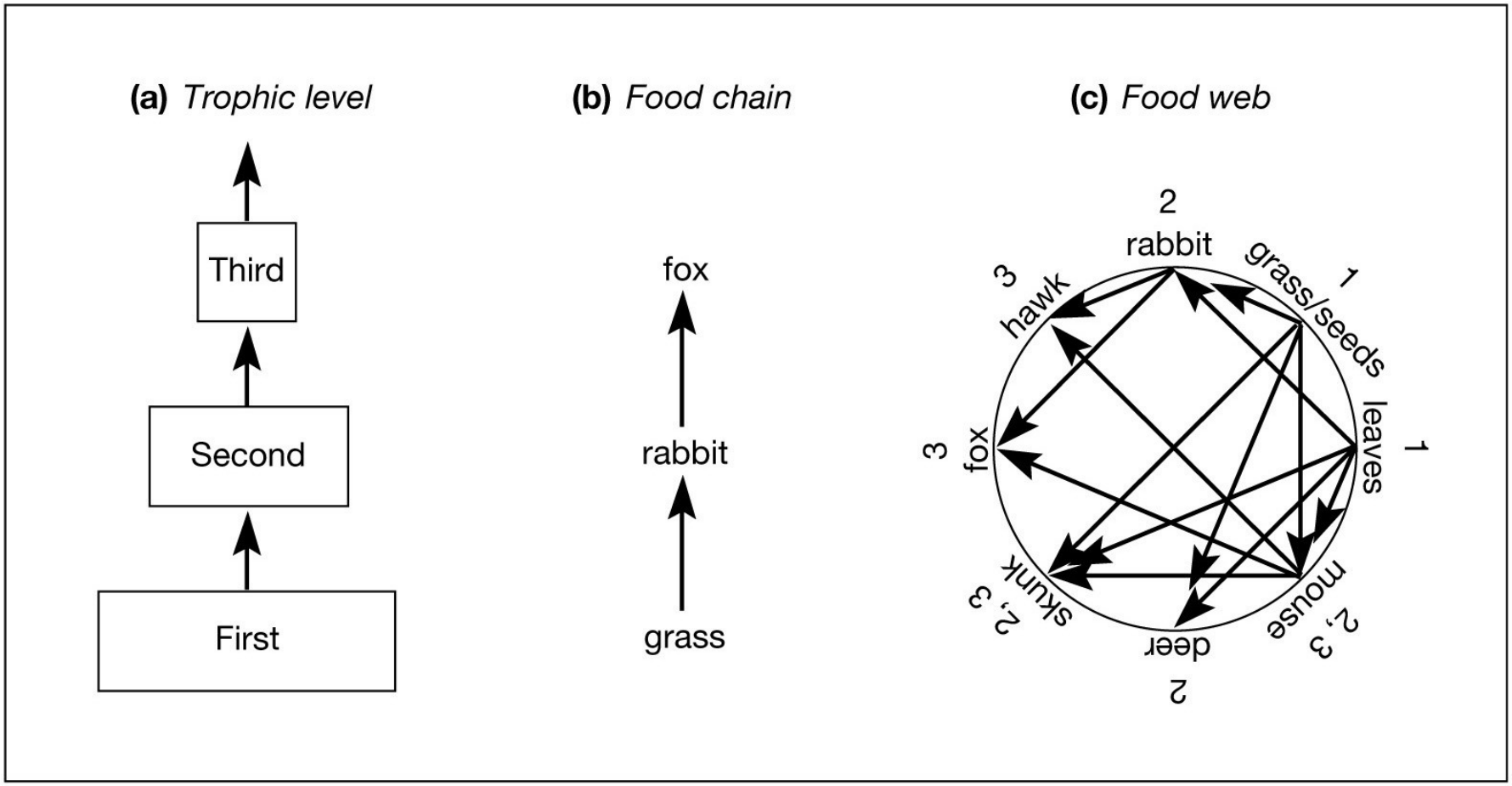
c Predicted large herbivore biodiversity
(% of continental species pool)



Ecological Pyramid



Energy and Food Transfers in Food Chains and Food Webs

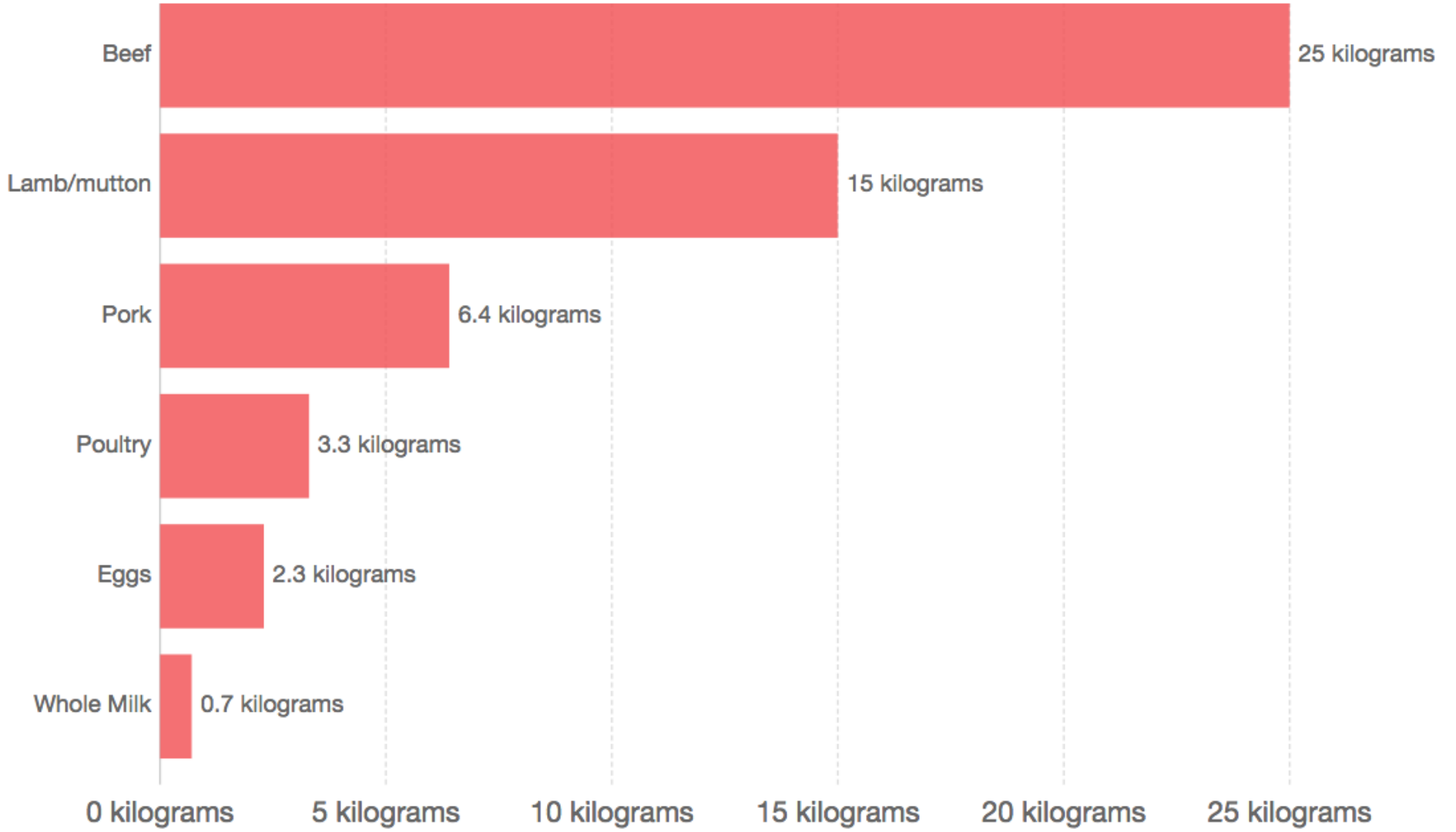


Energy transfer efficiency between trophic levels = 10-20%

Feed required to produce one kilogram of meat or dairy product

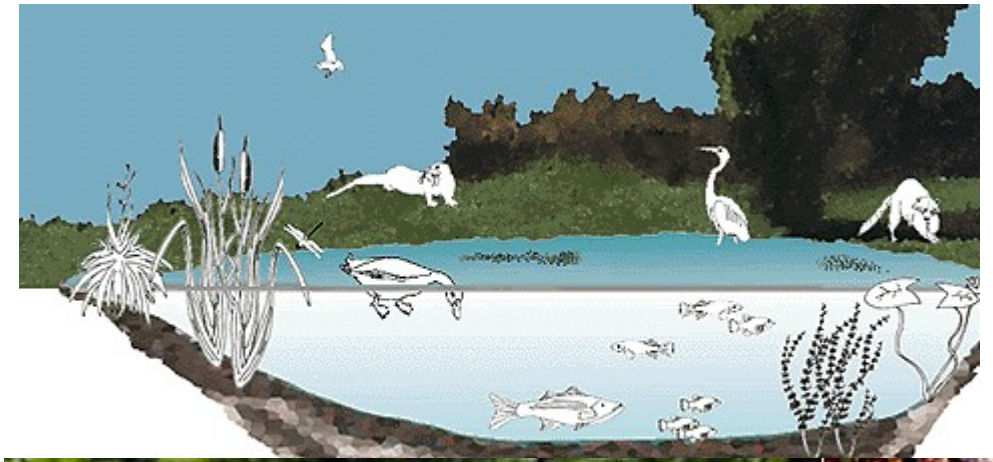
Our World in Data

Quantity of animal feed required to produce one kilogram of meat, egg or milk product. This is measured as dry matter feed in kilograms per kilogram of edible weight output.



Source: Alexander et al. (2016)

OurWorldInData.org/meat-and-seafood-production-consumption/ • CC BY-SA



Ecosystem services

Reading from the Ecological Society of America





Types of Ecosystem Services
here, examples from wetland ecosystems

Provisioning Services

Products obtained from wetland ecosystems:

- Food
- Fresh Water
- Fibre & Fuel
- Genetic resources
- Biochemical Products

Regulating Services

Benefits obtained from regulation of wetland ecosystem processes:

- Climate regulation
- Hydrological regimes
- Erosion Protection
- Reduction of Natural Hazard risk
- Pollution Control & Detoxification processes

Cultural Services

Material and non-material benefits obtained from wetland ecosystems:[1]

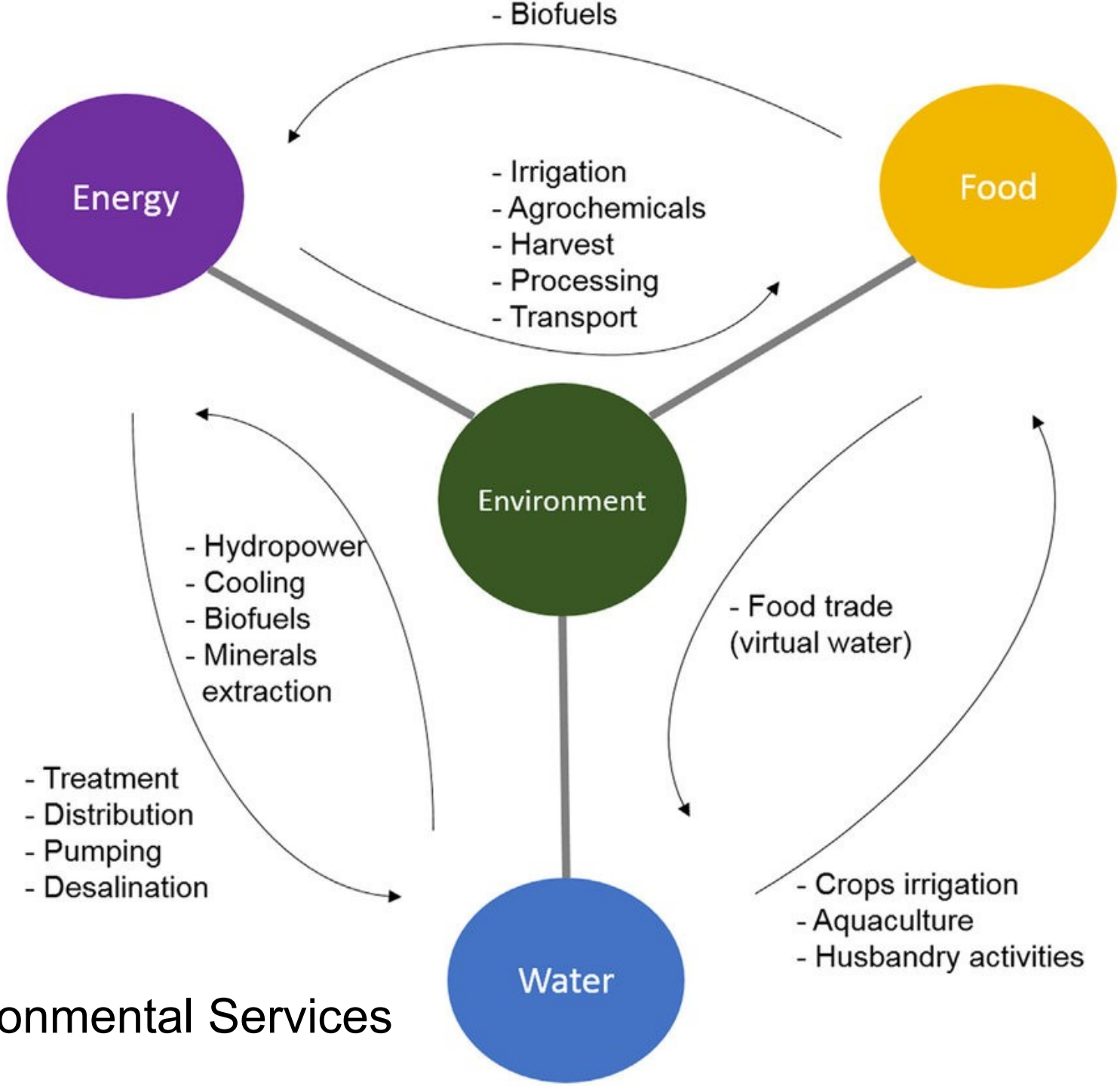
- Spiritual & Inspirational
- Recreational
- Aesthetic
- Educational
- Historical Artifacts
- Traditional Livelihoods & Knowledge

Supporting Services

Services necessary for the production of all other ecosystem services:

- Soil formation
- Nutrient cycling
- Primary production

[1] Note that "Cultural Services" includes material and non-material cultural values, benefits and functions (as identified in COP8 DOC.15. *Cultural aspects of wetlands*).



Environmental Services

Resilience and Regime Shift

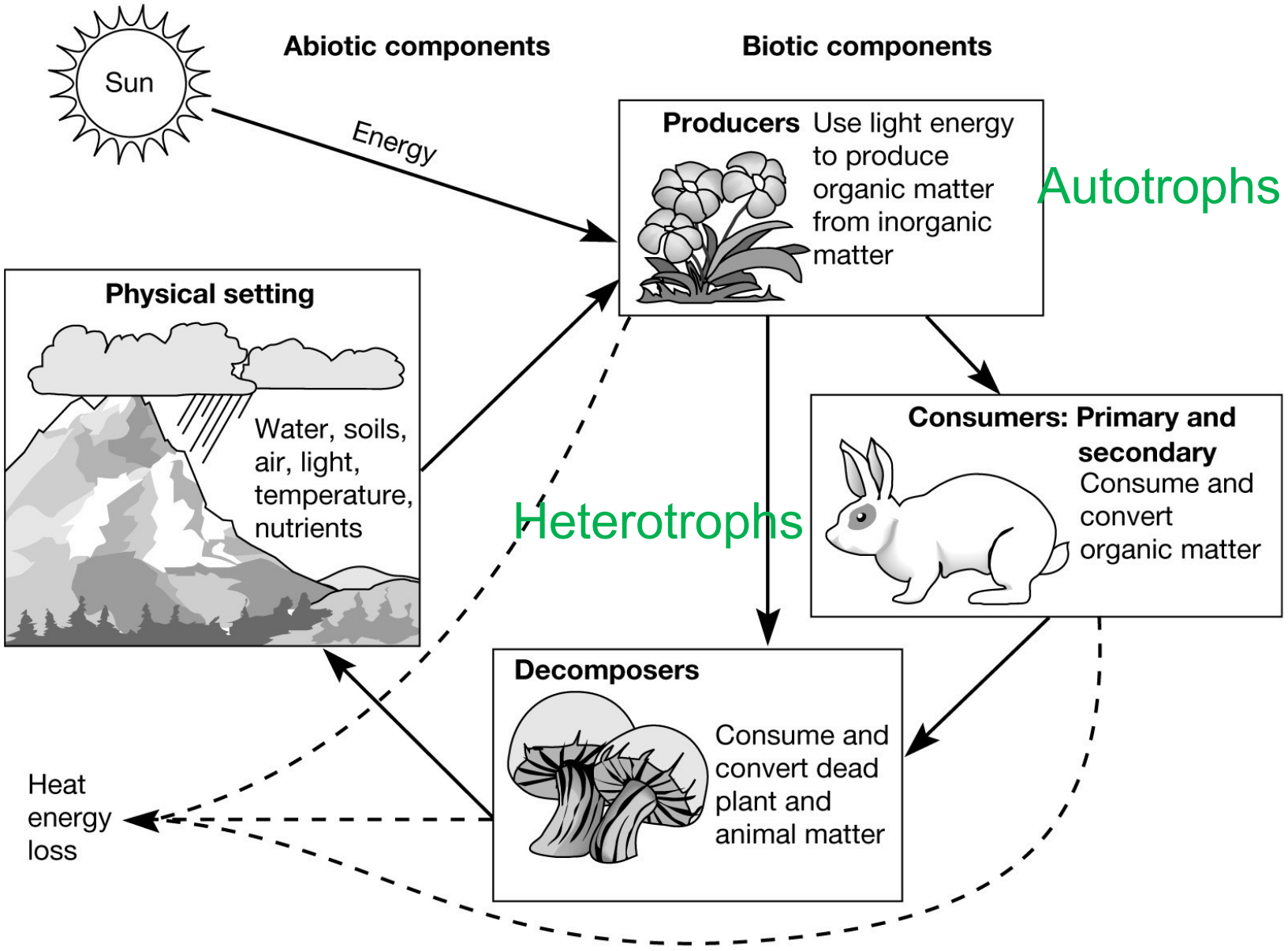
Folke et al., ”Regime shift, resilience…”,
In ‘Readings’ on myCourses

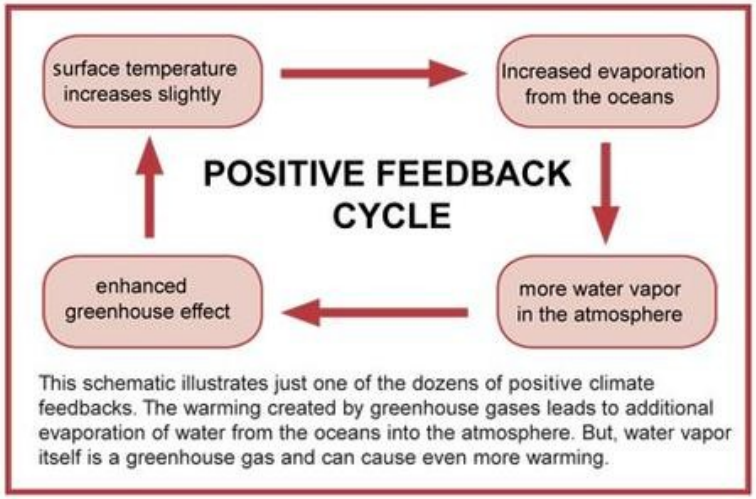
(this reading illustrates key concepts in this lecture: ecosystem regime shifts & resilience; applied to different ecosystems…)

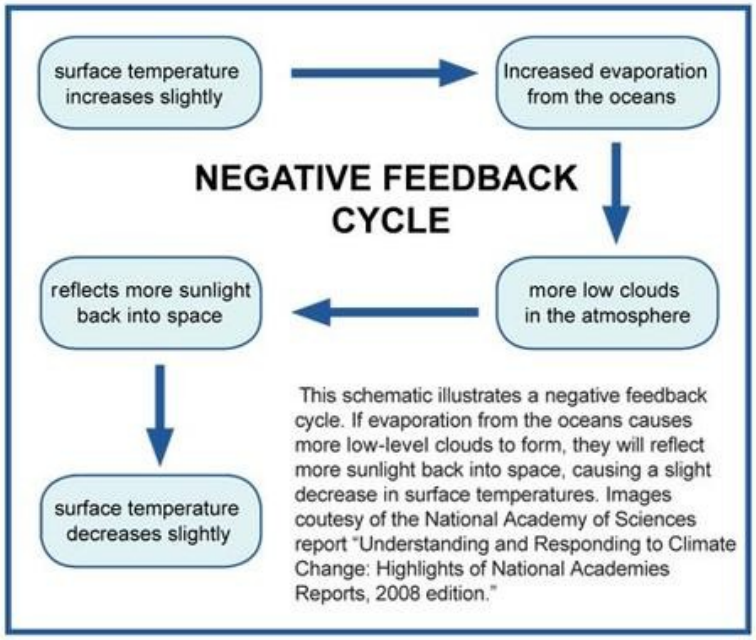
Concept of ‘Regime shifts’: sudden, large scale and hard to reverse transformations of a system, in response to slowly varying changes in some ‘driver(s)’

The role of human agents interacting with natural processes in triggering such ‘unexpected shifts’

Ecosystem Components: but Feedbacks

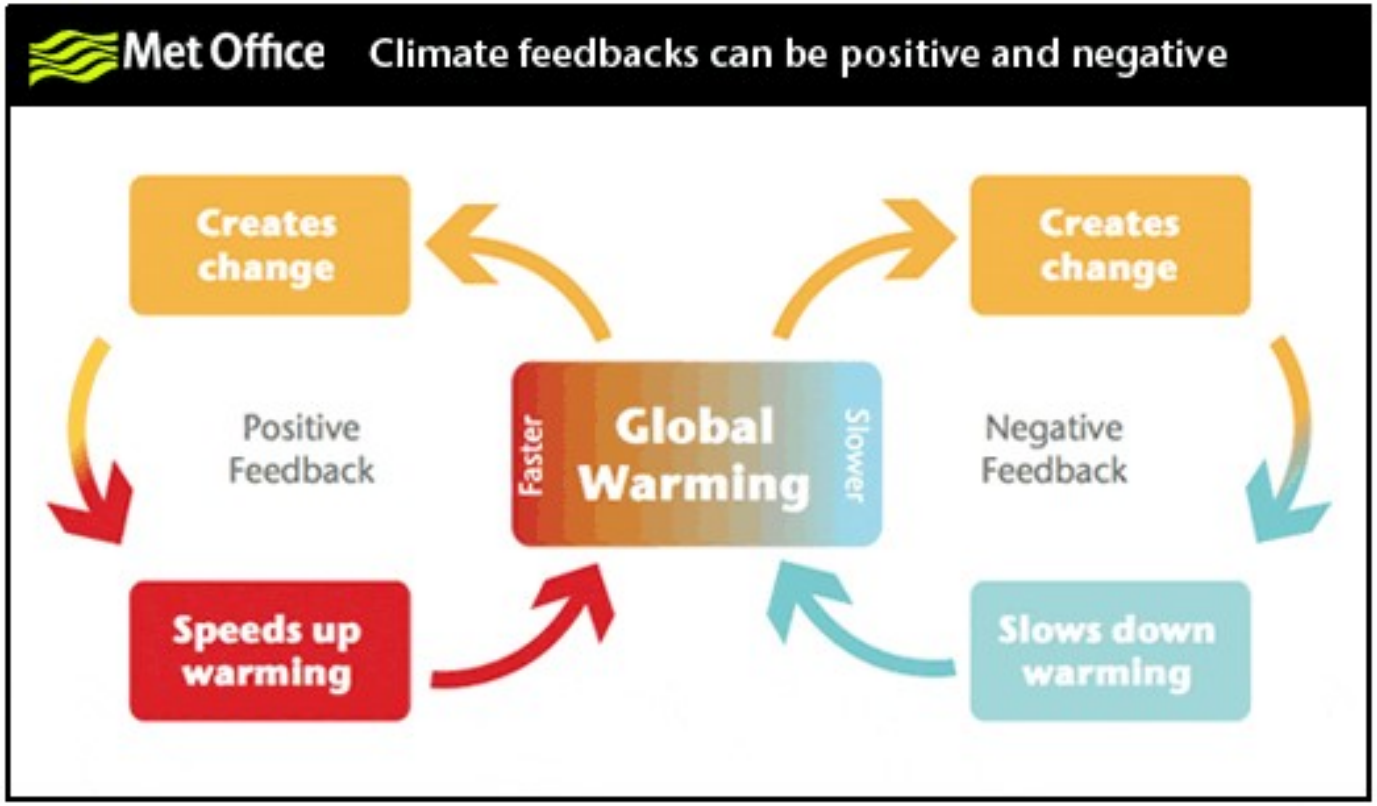




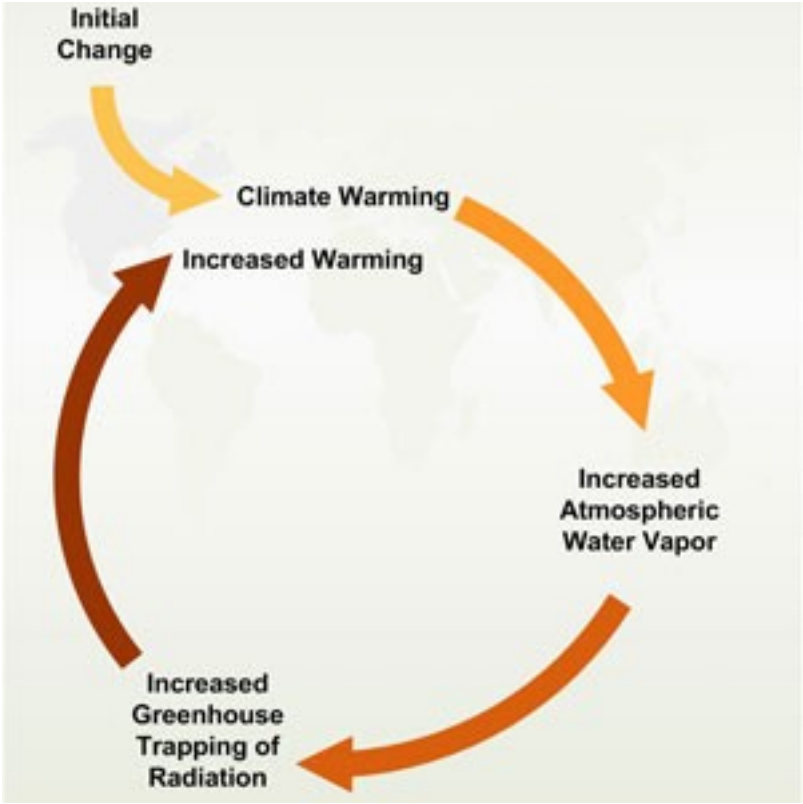


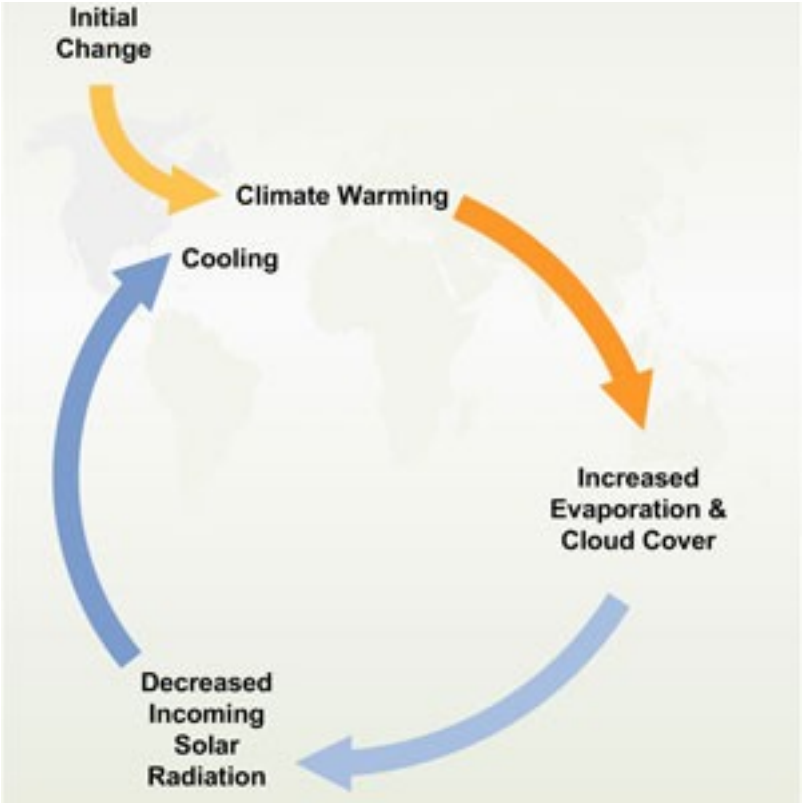
Feedback Example: Earths climate system: Sun insolation is an external driver, but the system state at any time involves feedback responses by oceans, land cover (albedo), human fossil fuel consumption, clouds, weathering of rocks , etc.

Stability of any system depends on the interaction between a number of positive and negative internal feedbacks

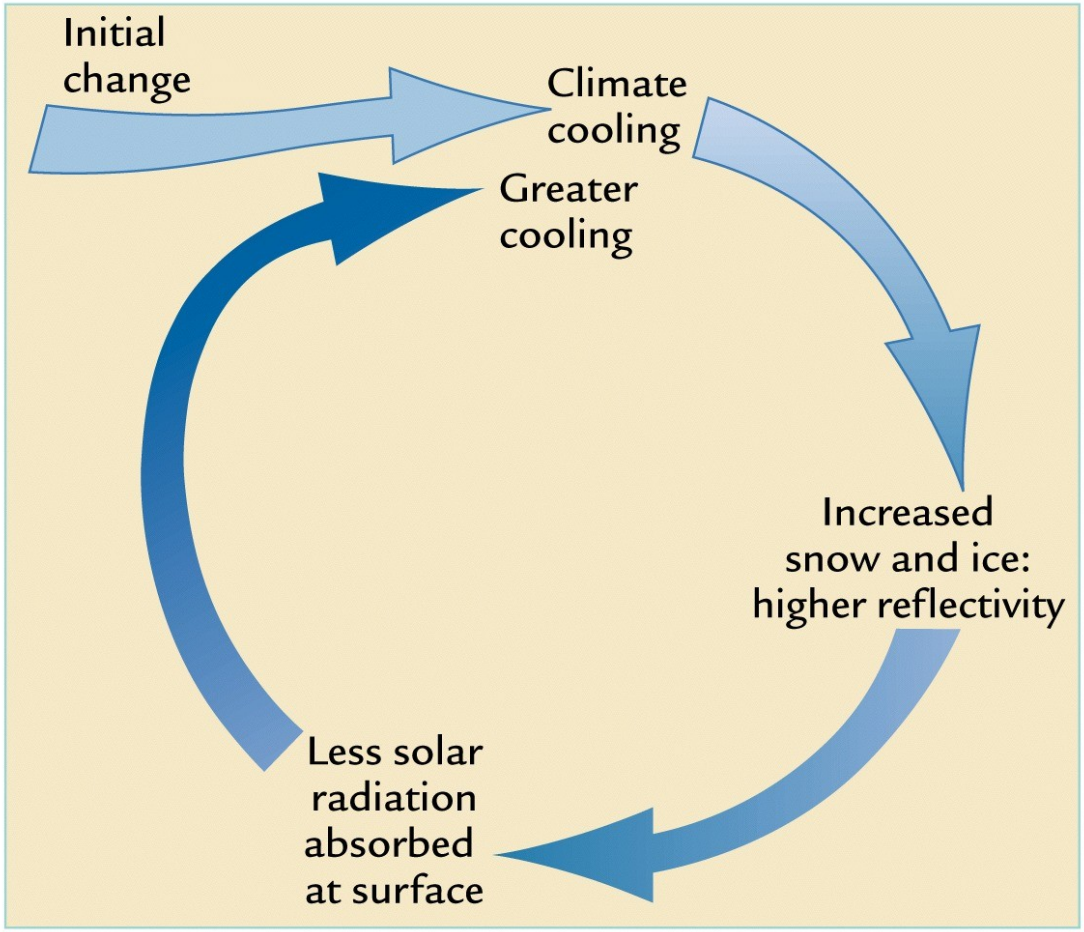


There are dozens of other + and – feedbacks loops operating in the climate system, some will be explained in coming lectures

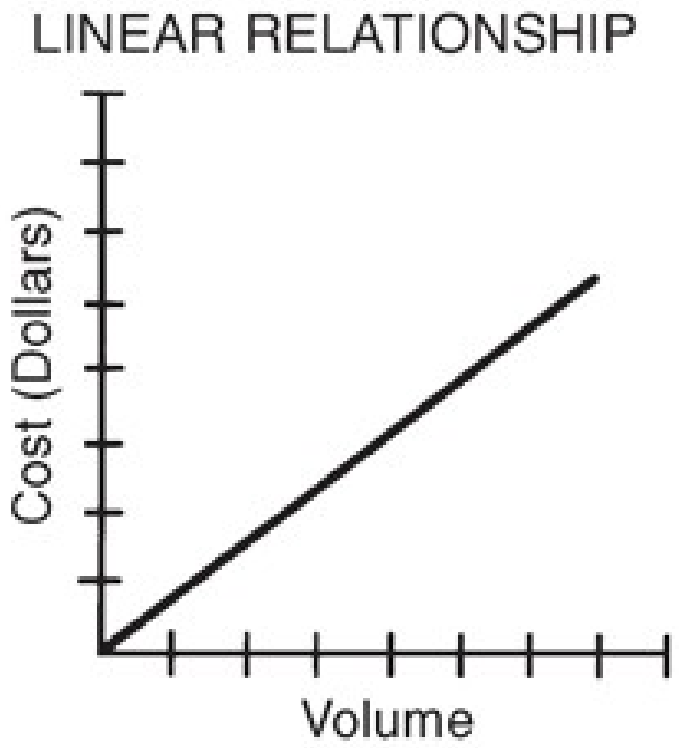


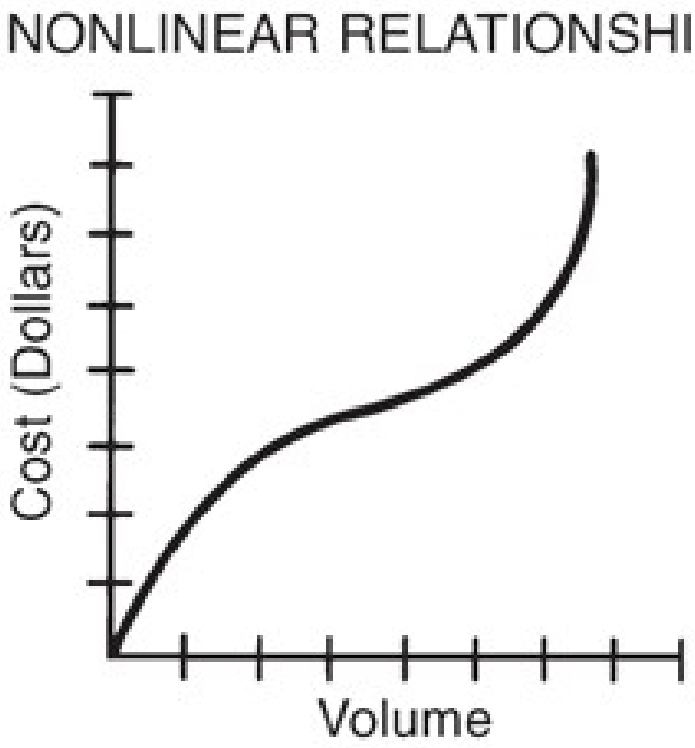


Many other types of feedbacks govern climate at different time scales



Real system relations are not necessarily “linear”.
Changes may kick in more strongly only at certain levels





Short TASK: This weekend (Monday before midnight)

Think of two feedbacks in your experience:

A Positive (reenforcing) Feedback

A Negative (diminishing, restoring) Feedback

Post them in 'Assignments' -we'll compile the early submissions and discuss them in class on Monday. Creativity is nice, but the feedback examples won't be graded –they are for discussion.

Some feedbacks discussed in class:
El Niño - La Niña southern oscillation
Ice-albedo feedback
Clouds (shortwave versus longwave behavior)

How stable are lake ecosystems? (how unchanging are there chemical and biological characteristics?)





Healthy lakes with good dissolved oxygen levels and diversity of fish species





No sudden change had occurred in the watershed, so that the ‘culprit’ driver of the change was long mysterious